

Cognitive Biology: The Mouse Model and the Attention-Head Manifold

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Abstract

This series computes a body from a genome as a neural cellular automaton whose per-cell rule is read by an outer genomic model, through four morphogenetic attention heads that differentiate, divide, migrate, and shape a field of cells. This paper takes that construction to the real mouse embryo and builds it out against the whole-embryo Stereo-seq atlas of organogenesis, the E9.5-to-E13.5 series. The method is a single working loop run in both directions. In the forward direction a quantity read from the atlas, an organ's position, its identity, the body's axes, the curl of its trunk, is asked whether it follows from the genome. In the reverse direction a quantity the genome gives, a Hox address, a master-transcription-factor cluster, an elongation program, is asked whether the atlas implements it. The heads are added or converted as each direction resolves. A head is a master-transcription-factor super-enhancer cluster, and the head methodology is a single bridge between the genome and the atlas: the SEdb super-enhancer atlas enumerates the candidate clusters the genome carries, the whole-embryo Stereo-seq atlas (MOSTA) annotates which of them a tissue expresses and names each by its most specific master regulon, and the AlphaGenome sequence model resolves the heads the atlas is too coarse to name. The central result is that, grown forward from a single cell, the frame and its heads produce a body a reader recognises on sight as a vertebrate: a head resolved into its fore-, mid-, and hind-brain regions, four limbs on the antinodes of the body's left-right electric eigenmode, the organs filling the antinode grid in the order the master clock unlocks them, the neural tube along the dorsal midline, and the axis tapering into a tail. The shape is the model's own, grown with no atlas silhouette supplied and no warping onto a real section, so the recognisable tetrapod is a prediction of the compiler and not a fit to the mouse. The compiler then gives the placed body its form. A small repertoire of gene-switched mechanical behaviours, the shape head, folds the trunk into the fetal curl and makes the individual organs: the heart's crescent fuses into a tube and loops to the right under the laterality cascade, the gut coils, the limbs grow out along their proximo-distal axis, and the optic vesicle cups. The placed anatomy thus becomes a folded embryo carrying a looped heart, a coiled gut, and grown limbs at their real positions. That these are the right shapes and not merely plausible ones is measured against a dense three-dimensional reconstruction of the real embryo, stacked from thirteen serial sections of the E16.5 fetus, at which density the internal topology a single section hides resolves. Every one of the six modelled organs matches the real more closely than a shapeless-cloud

null, the mean match rising from 0.47 to 0.63, and the heart, modelled as a thin-walled luminal tube, recovers the tortuosity of its loop to score 0.88 against the 0.52 of the cloud. Against the atlas the placement holds. The organ order and their antero-posterior positions are reproduced to within four percent of body length at a correlation of 0.99, each annotated tissue's most specific master transcription factor is its textbook regulator, and the low gap-junction eigenmodes are the antero-posterior and dorso-ventral axes at 0.84 and 0.80. Across the series the annotated head set resolves from progenitor fields into their differentiated derivative heads in the order the clock predicts: the neural crest into the dorsal-root ganglia, the jaw, the meninges, and the sympathetic chain, the sclerotome into the vertebral cartilage, and the urogenital ridge into the kidney and the gonad. The peripheral-nervous-system, cartilage, and vascular heads so revealed are added to the compiler and validated to unlock before their derivatives and to sit within the progenitor territory. The limbs are placed on the same electric-body frame that carries the face and the mammary line, and the entry of the left-right eigenmode into the operator's spectrum as the body gains medio-lateral width is identified as the geometric signature of the fish-to-tetrapod transition that produces the amphibians. Finally the heads are shown to be a coupled dynamical system. Each organ's master-transcription-factor signature is a trajectory in identity space that diverges from its neighbours as differentiation proceeds while riding a shared developmental tempo, its identity kept orthogonal to its position, and neighbouring heads pull one another along in proportion to their physical contact. The atlas attributes that coupling mainly to the mechanical fascia rather than the electrical channel, and a forward ablation confirms it causally, the removal of the notochord displacing the identity of the vertebral cartilage more than any other tissue. Because the outer genomic model emits the heads as a coupled set, its output lies on a low-rank manifold, measured on the mouse series to be of order seven dimensions rather than the twenty-seven of independent heads, so a species is a coordinated low-rank move on that manifold and an individual variant propagates through the coupling as pleiotropy. The developmental head-coupling matrix and the cross-head covariance of a genomic sequence model are two measurements of one manifold that agree on its dimensionality and its leading axes. And the same coupling makes a concrete prediction for loss-of-function genetics, that a single-gene knockout should disrupt the several organs its gene serves together, so the organ-by-organ co-disruption structure of a genome-wide knockout panel is the attention-head manifold read out in loss of function. An honest accounting closes the paper: the order, the identities, the axes, the antero-posterior calibration, and the curl are read from the genome or the atlas that expresses it, while the confinement that scales the curl, the anterior default of the cranial organs, and the bioelectric setpoints remain anchored, the last because a resting potential is a property of channel gating and not of transcript level.

1 The Genome, the Attention Heads, and the Mouse

The earlier papers in this series compute a body from a genome at the resolution of cells. A neural cellular automaton, whose per-cell rule is read from the genome by an outer large genomic model, grows a field of cells that differentiate, divide, migrate, and take shape through four morphogenetic attention heads [1, 4, 5]. What that construction has lacked is a sustained confrontation with a real vertebrate embryo across developmental time. This paper supplies it, using the whole-embryo spatial transcriptomic atlas of mouse organogenesis, five Stereo-seq sections from E9.5 to E13.5 [8], as the ground truth against which the compiler is both tested and extended.

The method is a single loop, run stage by stage and in both directions. In the forward direction a quantity is read from the atlas, an organ’s position, its identity, the body’s axes, the curl of its trunk, and it is asked whether that quantity follows from the genome. Where it does, the constant the model had carried is replaced by the genome read. Where it does not, the constant is kept and named as an anchor, in the sense in which the resting-potential scale of the foundational paper is an anchor, fitted now and derivable later. In the reverse direction a quantity the genome gives, a Hox address, a master-transcription-factor cluster, a convergent-extension elongation program, is taken to the atlas and asked whether the real embryo implements it. The output of the loop is head design: organ heads present in the atlas but absent from the compiler are added, and fitted anchors are converted to genome reads one at a time. The attention heads sit in the middle of this loop as the object that both sides constrain. An organ is its master-transcription-factor head, its identity a cluster of super-enhancers read from the accessibility code and its position an address on the body’s electric frame [3]. This fixes the head methodology that runs through the paper, a bridge between the genome and the atlas. On the genome side a head is a super-enhancer cluster, read by the series’ regulatory front-end: the activity-by-contact enhancer–promoter links that tie a super-enhancer to its master gene, the SEdb super-enhancer atlas that catalogues the clusters, and the AlphaGenome sequence model that predicts them from sequence [3, 10, 11]. On the atlas side a head is a tissue the whole-embryo Stereo-seq atlas annotates and whose most specific master-transcription-factor regulon names it. The methodology of this paper is that bridge, SEdb-to-head-to-MOSTA: the super-enhancer catalogue supplies the candidate heads the genome carries, the atlas says which of them a tissue expresses and with which master, and the sequence model resolves the heads the atlas is too coarse to name (Section 3.2).

Between the genome and the atlas, the result of the loop is a body. Grown forward from one cell, the frame and its heads produce a recognisable vertebrate whose shape is the model’s own. The bulk of what follows is the demonstration that this body has its organs in the right places, its identities measured rather than asserted, its head set resolving in the order the clock predicts, and its heads coupled into one integrated system. That coupling proves to be more than a developmental fact: it places the heads on a low-rank manifold, the second arc of the paper alongside the mouse model. And it makes a concrete prediction for genetics, that a gene serving several heads disrupts them together, so the manifold is what a knockout panel should read out in loss of function.¹

Section 2 places the limbs on the electric-body eigenmode frame and identifies the appearance of the left-right eigenmode with body width as the fish-to-tetrapod threshold. Section 3 generalises the frame from the limb to the whole anatomy, shows the vertebrate the compiler grows, and validates the placement, the identities, and the axes against the mouse atlas. It then traces the head set resolving from progenitor fields into differentiated derivative heads across the series, and gives the honest accounting of what is computed and what stays anchored. Section 4 shows the heads to be a

¹This tissue-and-organ account connects back to the cell-resolution compiler of the earlier papers by a coarse-graining built to be invertible. A restriction operator $\mathcal{R}$ averages the cells of a part into its named coarse form, and a lifting operator $\mathcal{P}$ regrows a cell population inside a part’s boundary; because one genome parameterises both scales the round trip is consistent by construction, $\mathcal{R}(\mathcal{P}(\text{part})) \approx \text{part}$, which we verify on a bone to below 0.1% in length. Under the same coarse-graining each of the four cell heads maps to an organ head of the same functional form with renormalised parameters, so that an organ is a first-class operator and not a relabelling of cells. This is the developmental instance of a heterogeneous multiscale method whose coarse closure is genomic [6, 7]: it lets the coarse anatomy be refined back to cells when compute allows. The machinery is not needed for the results below and is not developed here.

coupled system of drifting identities that pull one another through their shared boundary. It draws the consequence that the outer genomic model emits them on a low-rank manifold, which makes a species a coordinated low-rank move and an individual variant a pleiotropy, and notes that the coupling predicts the organ co-disruption a genome-wide knockout panel measures. Section 5 adds the morphogenetic programs that give the placed anatomy its shape, folding the trunk into the fetal curl and forming the individual organs, the heart looping, the gut coiling, the limbs growing out. It scores those forms against a dense three-dimensional reconstruction of the E16.5 fetus, where the internal topology a single section cannot show resolves. Section 6 points, without developing it, at surgery-free operations.

2 The Origin of the Limb: the Electric-Body Frame and the Amphibian Threshold

The limbs are a special case of a general rule, and stating it makes a claim about their evolutionary origin. A limb is a bilateral, paired appendage placed at a definite antero-posterior level, and a tetrapod has two such levels, the fore-limb and the hind-limb. In this framework a repeated, bilaterally symmetric feature is not placed by coordinates. It is placed on the antinodes of the low eigenmodes of the tissue's own gap-junction operator, the electric-body frame, exactly as the facial features, the mammary line, and the abdominal muscles are placed [3]. The construction has two ingredients. The antero-posterior eigenmode is the head-to-tail coordinate, and the Hox code sets the fore- and hind-limb levels along it. The left-right eigenmode is the bilateral frame: its nodal surface is the midline and its two antinodes are the left and right flanks. A limb bud forms where an antero-posterior level meets a left-right antinode, and the pair is symmetric by construction, because the two antinodes are mirror images about the mode's node. Nothing in this is specific to limbs. It is the same frame, read at two Hox levels, that the earlier work read along the milk line for the mammary glands and along the trunk for the abdominal muscles.

The consequence is a geometric precondition for having limbs at all. The left-right eigenmode exists in the usable, low part of the operator's spectrum only if the body has real medio-lateral width. A body drawn out by convergent extension into a thin, axially elongated form has, among its low modes, only the antero-posterior and dorso-ventral axes. The eigenvalue of the first left-right mode scales as the inverse square of the width, so in a narrow body that mode is pushed high, out of reach, and a bilateral pair has no frame to sit on. Such a body is limbless. As the body broadens, the left-right eigenvalue falls until the mode enters the low band, and only then can bilateral limbs be placed on its antinodes.

We measure this on the grown embryo (Figure 1). The antero-posterior eigenmode recovers the head-to-tail axis at a correlation of 0.94. In the thin, collapsed body the best correlation of any low mode with the left-right axis is 0.02: there is no left-right mode, and no bilateral limb forms. As the medio-lateral width is increased, that correlation holds near zero and then rises sharply, through a threshold at about half the full width, to 0.96. At that point the left-right mode is a clean bilateral field with its node on the midline, and the four limb buds sit on its four antinode intersections.

This threshold is, in the language of the compiler, the developmental signature of the fish-to-tetrapod transition, and it locates where the amphibian body plan is born. A narrow, axially elongated, finned body has no left-right frame and therefore no paired limbs. The broadening of

the body that accompanies the origin of tetrapods is exactly what brings the left-right eigenmode into the spectrum and licenses the fore- and hind-limbs on its antinodes. The point is that this is not achieved by adding a limb gene. The paired-appendage genetic programme is shared between the fin and the limb, a deep homology. What changes at the threshold is not the genes but the geometry of the body wall, and through it the spectrum of the operator that positions the genes' output. In this account the amphibians are made not by a new instruction but by a change in shape that changes the frame.

The scope is the schematic embryo of the companion papers, and two limits are stated plainly. The threshold is demonstrated as a spectral fact of the tissue operator under a controlled widening, and it is not fitted to the timing of the fossil record. And the width itself is set here by the convergent-extension parameter, a proxy for the real developmental broadening of the body wall. What is robust is the qualitative claim: a left-right eigenmode is a geometric prerequisite for a bilateral limb, and it is width that summons it.

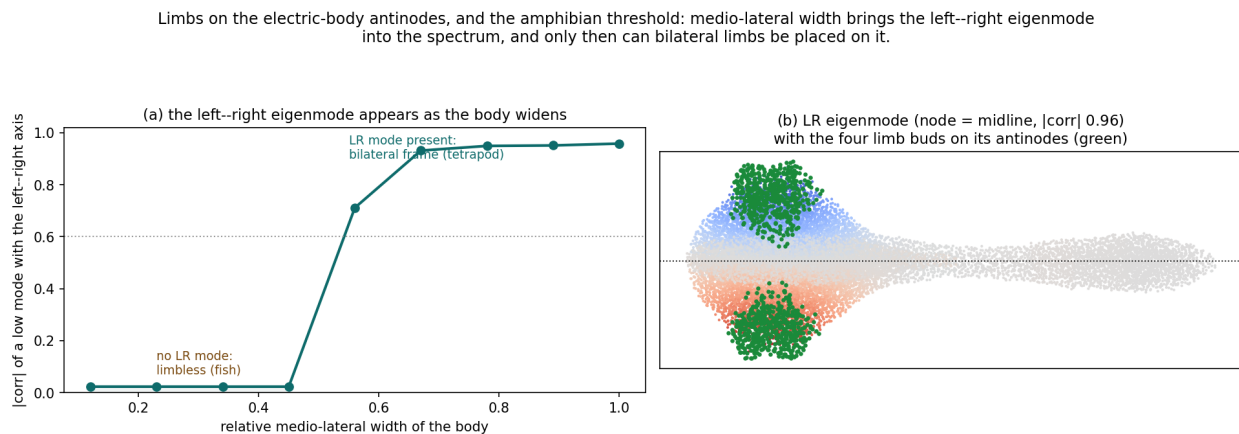


Figure 1: Limbs on the electric-body antinodes, and the amphibian threshold. Left: the best correlation of a low gap-junction eigenmode with the left-right axis, as the body is widened in the medio-lateral direction; it stays near zero for a thin body (no left-right mode, a limbless, fish-like body) and rises sharply through a threshold at about half the full width to 0.96 (the left-right mode present, a bilateral frame, tetrapod). Right: at full width, the left-right eigenmode (top view, coloured by mode value, its node on the midline) with the four limb buds placed on its antinodes (green). The antero-posterior eigenmode supplies the coordinate on which Hox sets the fore- and hind-limb levels; the left-right eigenmode supplies the two bilateral sides. The transition is a change in body geometry that changes the operator's spectrum, not a change in the genes.

3 From the Limb to the Whole Anatomy, and its Validation against a Real Atlas

The limb is the clearest case of a rule that places the whole anatomy. Every repeated or bilaterally definite structure sits on an antinode of the body's electric frame, the low eigenmodes of its own gap-junction operator. An organ is placed no differently from a limb: a midline organ, the heart,

the liver, the gut, on the nodal surface of the left-right mode, and a paired organ, the eye, the otic vesicle, the lung, the kidney, on its two antinodes, each at an antero-posterior level picked out by an antinode of the head-to-tail mode. The organs fill this antinode grid in the order the master clock unlocks them, the telomere-gated withdrawal of PRC2 that reads the differentiation strata. Lateral inhibition enforces one structure per antinode, so that no two organs occupy the same locus [5]. In the grown embryo the order in which the organs unlock matches the antero-posterior order of the antinodes they take, at a rank correlation of 0.997. The antinode frame predicts an organ's position better than the bioelectric potential itself, which is the readout of the frame rather than its cause.

The organs are the leaves of an attention-head tree whose depth is developmental time (Figure 3): at the root the axis heads that make the body's length, width, and electric frame; in the middle a shared library of effector heads, adhesion, migration, apoptosis, fusion, folding, and lateral inhibition, that any organ calls; and at the leaves one organ head per super-enhancer cluster, each a subroutine that invokes the shared effectors at an address on the frame [3]. Some heads are atomic, and some are progenitor fields that resolve into a set of subheads, the neural crest into its ganglia, jaw, and meninges, the neural tube into its brain regions, so that the tree carries a subhead layer beneath the organ leaves. Two of the heads make the logic explicit. The notochord is the left-right node made physical, the axial organiser rod laid down on the midline where the limb sat on the flanks, and its master transcription factor is Brachyury. The skin is the epidermal envelope, the outer shell over the whole body. The ribs and the axial musculature are the segmented case of the same law, not a single antinode but one bilateral block per somite, their periodicity inherited from the segmentation clock rather than imposed, the metamerism reading of the frame that the eyes and the limbs read once [5].

The effector library is called selectively, and lateral inhibition is the clearest instance of an effector that only some heads need. Lateral inhibition makes a structure discrete and spaced, so it is required by the repeated and the paired organs, the two eyes and otic vesicles on the two antinodes of the left-right mode and the segmental somites, ribs, and dorsal-root ganglia on the periodic antinodes of the clock, where it fixes how many there are and how far apart. The single asymmetric viscera call no such rule, because there is only one of each to space: the looped heart, the liver held to the right, the spleen and stomach to the left, the coiling gut. These are placed and composed by the wave operator itself, the left-right standing mode whose node is the midline together with the symmetry-breaking that biases the mode to one side, the Nodal-to-Pitx2 cascade of laterality. The asymmetric composition of the trunk is thus a standing-wave phenomenon read off the eigenmode amplitude, not a spacing rule. The wave operator places, and lateral inhibition discretises only where a structure repeats.

The recursion does not stop at the organ. An organ is a head, and a head at a smaller scale carries the same two constructions the body carries. It has a surface-covering subhead, the organ-scale counterpart of the epidermis, called to the organ's outside by the same rule that calls the epidermis to the body's: the epicardium over the heart, the capsule over the liver and the kidney, the serosa over the gut, and the meninges over the central nervous system, which is the instance the model already builds. And it has its own electric scaffold, the low eigenmodes of the organ's own gap-junction operator, whose conductance its super-enhancer cluster sets. So the master-transcription-factor head that gives the organ its identity also, through the connexins and channels it expresses, sets the operator whose modes are the organ's shape axes. This is not a conjecture at the operator level: the earlier organ-formation work found each organ's bioelectric target to be low-rank in the cymatic modes of its own conductance-weighted operator, the heart's left ventricle

resolving onto its apex-to-base activation mode and the gut tube onto its oral-to-aboral peristaltic mode, so that the conductance profile is literally the cymatic generator [3]. The prediction the recursion makes is then exact: an organ's subheads, the cardiac chambers, the regions of the brain, the zones of the kidney, should sit on the antinodes of the organ's own electric scaffold as the organs sit on the antinodes of the body's, a single machinery applied at body, organ, and subhead scale. We test this on the grown organs (Figure 2): building each organ's own gap-junction eigenmodes from its cells, its subheads are found to separate along them far above a label-shuffled null. The brain's four neuromeric regions, the fore-, mid-, and hind-brain and the cerebellum, fall on the antinodes of the brain's own first eigenmode, its antero-posterior axis, with a between-subhead variance fraction of 0.88. The cardiac chambers, the cortico-medullary zones of the kidney, and the two layers of the optic cup each separate on a low mode of their own organ at a significance of $p < 0.005$ against the null. The recursion is thus a measured fact and not only a prediction: the organ's subheads do sit on the antinodes of the organ's own electric scaffold, the same machinery one level down. The generative step follows directly. Marking each subhead's antinode locus in the organ's mode-space, the organ-scale analogue of a body antinode, and assigning every cell to the nearest antinode, the placement recovers the true subheads at 74 to 90% against a nearest-centroid chance of 27 to 50%: the brain's four neuromeres at 87%, the kidney's cortico-medullary zones and the optic cup's two layers near 90%, and the four cardiac chambers, the hardest case, at 74%. A subhead is therefore placed the way the body places an organ, on the antinodes of the electric frame one scale down, so the construction is generative and self-similar from the body through the organ to the subhead. The one thing the compiler still does by hand here is form the subhead pools, which it splits by geometric position. The electric placement of those pools, shown to recover them, is the route to doing that too by the frame alone.

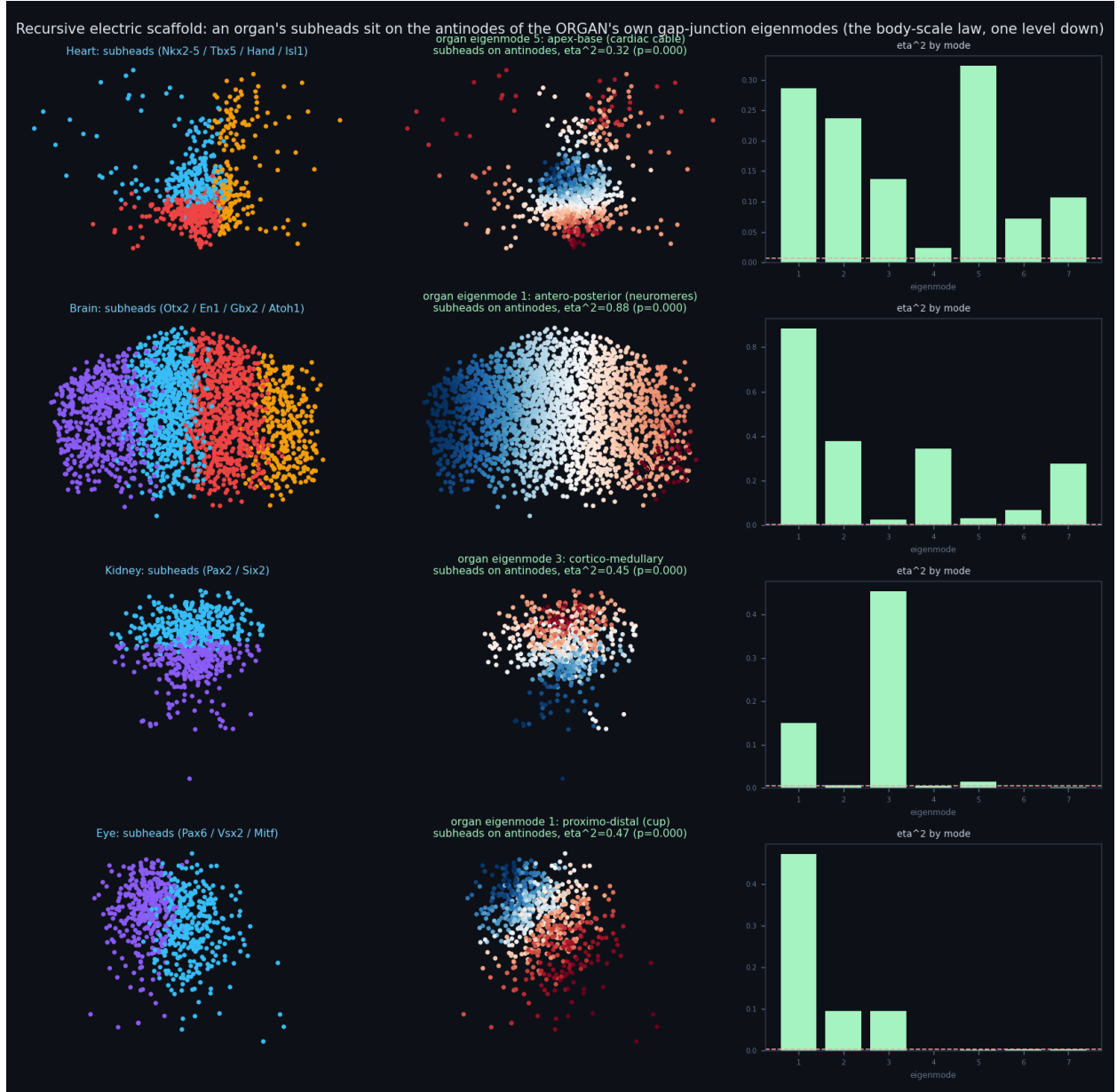
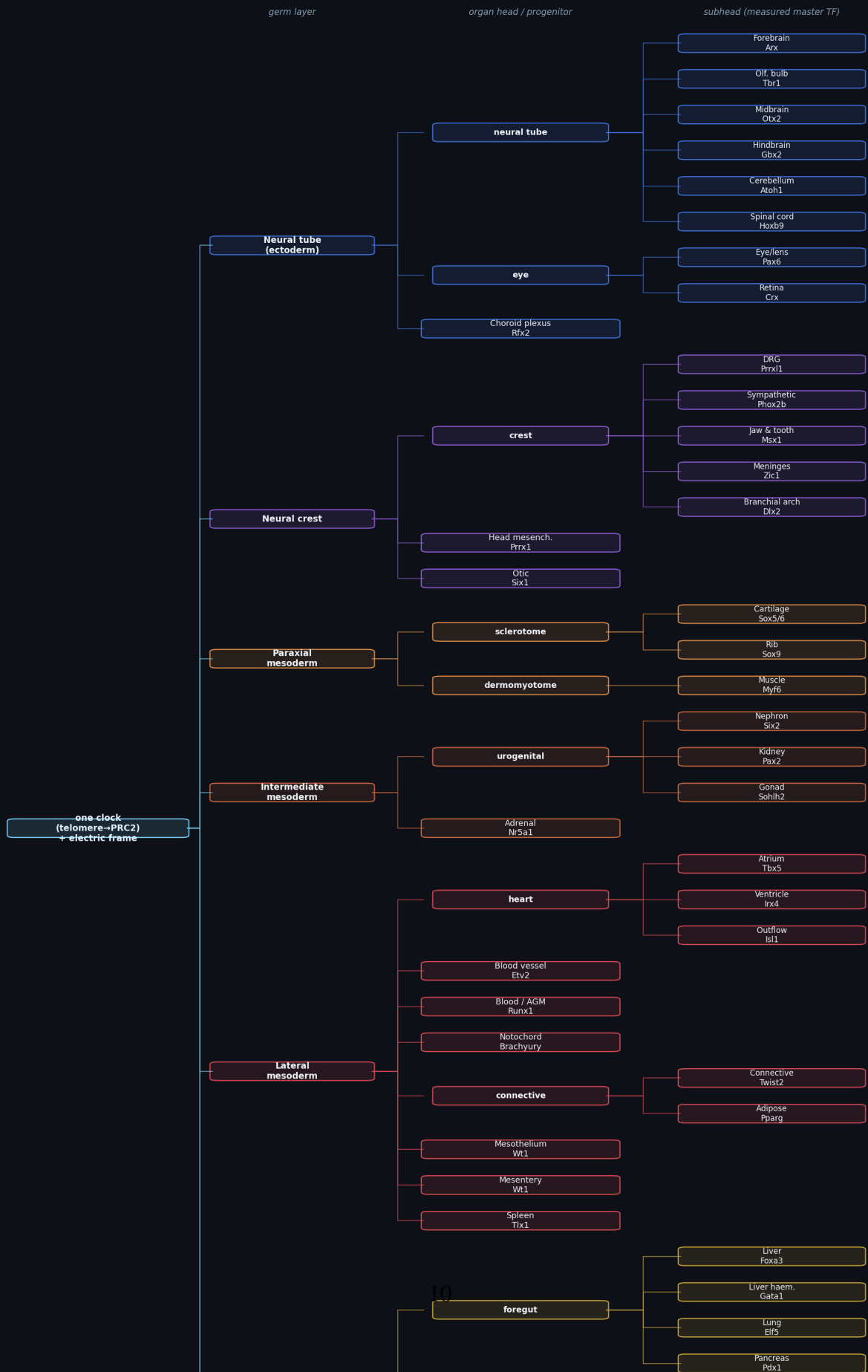


Figure 2: The recursive electric scaffold. For each organ (rows: heart, brain, kidney, eye), left, the organ's cells coloured by subhead; middle, the same cells coloured by the organ's own gap-junction eigenmode that best carries them; right, the between-subhead variance fraction (η^2) of each low eigenmode, with the label-shuffled null (dashed). The subheads sit on the antinodes of the organ's own modes as the organs sit on the body's: the brain's fore-, mid-, and hind-brain and cerebellum on the antinodes of the brain's first eigenmode, its antero-posterior axis ($\eta^2 = 0.88$), and the cardiac chambers, kidney zones, and optic-cup layers each on a low mode of their organ, every case significant at $p < 0.005$ against the null. The construction is the body-scale law applied one level down.

The face is the sharpest illustration of the recursion, because there the antinodes of the smaller

frame are morphogen primordia already named in the literature. The facial prominences, the frontonasal, maxillary, and mandibular swellings, are not a separate patterning layer laid over the face but the morphogen sources of the craniofacial scaffold itself, the frontonasal ectodermal zone with its Sonic-hedgehog and fibroblast-growth-factor boundary and the bone-morphogenetic-protein gradient across the prominences, which set the shape of the face and, in the birds, of the beak. In the recursion these prominences sit on the antinodes of the face's own electric operator exactly as the organ primordia sit on the antinodes of the body's. So setting the electric face and placing the prominences are one operation and not two, and the morphogens the prominences emit are precisely what the cellular automaton reads to grow the face forward. The earlier volumetric-morphogenesis program, which grows a face forward from morphogen primordia on a genome-derived gap-junction operator, is then this same construction recognised at the scale of the face, and the primordia and the electric frame are two readings of one prepattern. The prediction it makes is causal and testable, that the electric antinode is set before the frontonasal zone switches on its Sonic-hedgehog and fibroblast-growth-factor at that locus, which developmental voltage imaging across the prominence-forming window would decide.

The head tree: organ heads and the subheads that resolve from them



Grown forward from a single cell, the frame and its heads produce a body that a reader recognises on sight as a vertebrate (Figure 4): a head at one end resolved into its fore-, mid-, and hind-brain regions along the axis, four limbs on the antinodes of the left-right eigenmode, the organs filling the antinode grid in the order the clock unlocks them, the neural tube along the dorsal midline, the skin envelope over the whole, and the axis tapering into a tail. The shape is the model's own. The embryo is grown from one cell with only the convergent-extension width knob set, with no atlas silhouette supplied and no warping onto a real section, so the recognisable tetrapod is a prediction of the compiler and not a fit to the mouse. That a set of mechanisms each grounded independently, on different tissue and different data, should compose into a recognisable vertebrate by coincidence is remote. A body plan that was never fitted yet emerges anyway is the evidence that the compositional structure has captured the way a body is built rather than been shaped to resemble one.

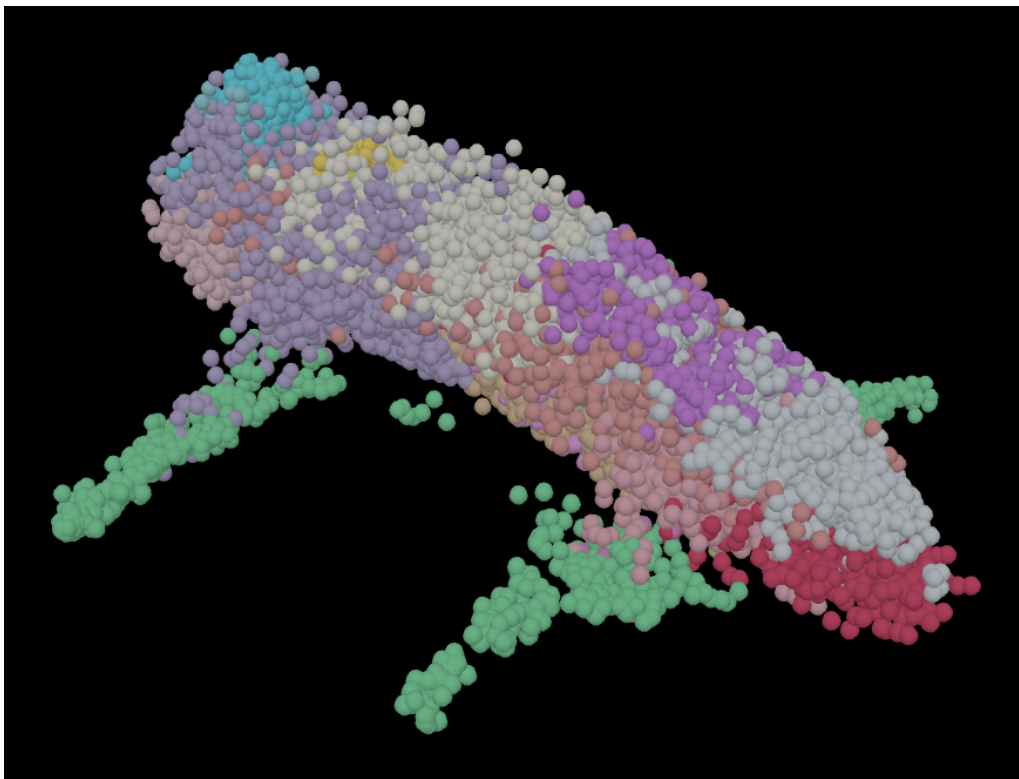


Figure 4: The vertebrate grown from a single cell by the compiler, in the model's own geometry. Each cell is coloured by its genome fate: the neural tube and its brain regions along the dorsal midline in blue, the eye at the head in cyan, the four limb buds on the flanks in green, and the trunk organs, the vertebral column, and the tail through the body. The body silhouette is intrinsic and not fitted: no atlas outline is supplied and the section is not warped onto a real embryo, and the only input that touches its shape is the convergent-extension width knob, the same genomic knob that sets the fish-to-tetrapod threshold. The organ-placement comparison against the atlas is made separately, on the real atlas outline, in Figure 7. The same body in lateral and dorsal view is shown in Figure 5.

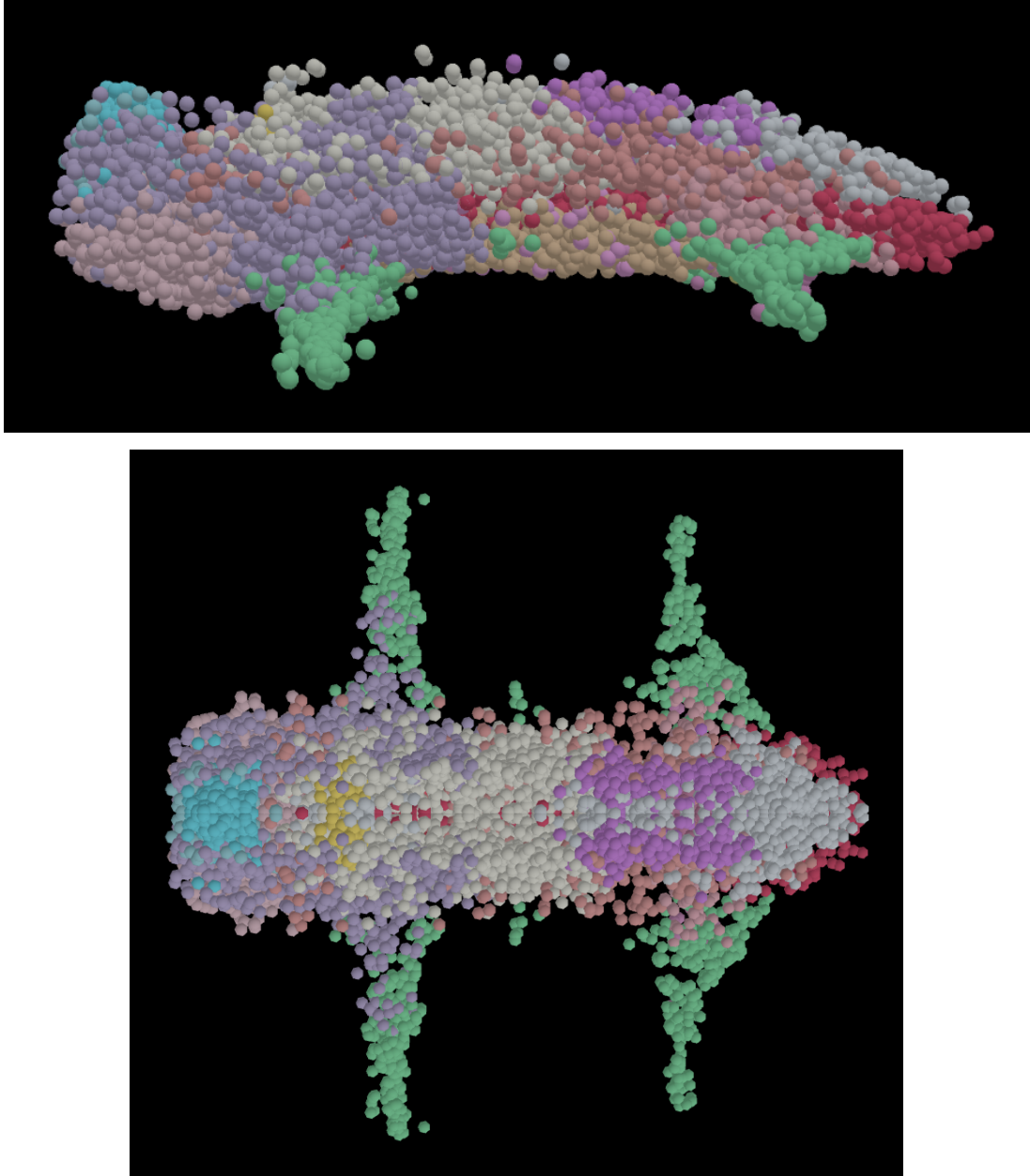


Figure 5: The same grown vertebrate in two orthogonal views. Top, the lateral (side) view: the head to the left with the eye, the neural tube along the dorsal edge, the limbs projecting ventrally, and the axis tapering to a tail at the right. Bottom, the dorsal (top) view: the four limbs splay symmetrically to the left and right about the midline, the fore-limb pair and the hind-limb pair, which is the signature of the bilateral tetrapod body plan, each limb seated on an antinode of the left-right eigenmode whose node is the midline the body is drawn about.

The antero-posterior levels are read from the genome rather than typed in. The methylation fossil record of a lineage, the hypomethylated enhancer archive it retains for life, holds the posterior Hox enhancers in its deepest, embryonic stratum, and the fore- and hind-limb levels are taken from the positions of the Hox9 and Hox10 enhancers in that archive [9]. The tissue is held together

by the two mechanical systems a real tissue has, a same-fate cadherin adhesion that sorts the compartments and a non-selective integrin-and-matrix fascia that binds them into one continuum. Making the fascia the longer-range and stronger of the two, as it is, is what keeps an out-growing limb bound to the body rather than tearing away, and is the difference between an embryo that fragments and one that holds.

We test the frame against the real mouse embryo, the E9.5 Stereo-seq atlas of the whole section [8], and four independent checks agree (Table 1). The organ positions come first: the antero-posterior levels at which the electric antinodes sit reproduce the real organ order and the absolute positions to within four percent of the body length, at a linear correlation of 0.99. The identities come second: the master-transcription-factor regulon most specific to each annotated tissue in the atlas is the textbook master of that organ, *Tbx20* for the heart, *Pax1* for the sclerotome, *Hnf4* for the liver, and *Brachyury* for the notochord, so that a head is a master-TF cluster read from the data and not asserted. The axes come third: the low eigenmodes of the embryo's own gap-junction operator, built on the real cell positions, are the body axes, the first the antero-posterior axis at a correlation of 0.84 and the second the dorso-ventral axis at 0.80. Weighting the operator by the measured connexin field returns the same axes as the geometry alone, because the connexins are broadly expressed, so the geometry sets the modes while the connexins supply the coupling that makes it a gap-junction operator at all. The axial code comes fourth: the posterior Hox colinearity is present in the real expression once the curved embryo is unbent into the straight frame on which the code was laid down. The unbending is not a fitted backbone but the intrinsic axis of the tissue itself, the Fiedler vector of the spatial gap-junction graph, the first non-trivial eigenvector of its Laplacian, which threads the body along its own curve where a straight principal axis cuts a chord across the arc. Read along this geodesic axis the colinearity of the Hox paralogs against their body level rises from 0.58 on the principal axis to 0.86, so the curvature was hiding the axial code and the tissue's own electric geometry recovers it.

Two limits are stated plainly. The comparison is to a single developmental stage and a schematic synthetic embryo, and the antinode ladder is coarse in the crowded mid-trunk, where the liver, the pancreas, and the lung fall on one antinode, so the secure claims are the order, the identities, and the axes, and not a fine metric map. And the organs are specified early, on the still-straight body, so the E9.5 atlas is a later, folded readout. This is why the axial reads require the unbending, and why the same code reads more cleanly on the zebrafish, which stays straight through organogenesis. What is secured is that the anatomy this frame places, in its order, its identities, and its axes, is the anatomy the real embryo has.

Organ	MOSTA region	MOSTA a	Antinode a	Δa	LR seat
Eye	Brain	0.09	0.14	0.05	antinode
Heart	Heart	0.35	0.39	0.04	node
Skin	Surface ectoderm	0.56	0.54	0.02	envelope
Rib	Sclerotome	0.66	0.69	0.03	antinode (segmented)
Kidney	AGM	0.72	0.69	0.04	antinode
Pancreas	Pancreas	0.75	0.79	0.04	node
Liver	Liver	0.81	0.79	0.02	node
Lung	Lung	0.85	0.79	0.06	antinode
Pearson $r = 0.99$, Spearman $\rho = 0.95$, mean $ \Delta a = 0.04$					

Table 1: The genome-derived electric-body antinodes against the real E9.5 mouse atlas (MOSTA). For each organ the atlas position along the antero-posterior axis a , in a body-length-normalised and unbent frame, is placed beside the nearest antinode of the grown embryo’s gap-junction operator, with the residual Δa and the left-right seat the frame assigns, a midline organ on the node and a paired organ on an antinode. The antinodes reproduce the real order and the positions to within four percent of body length.

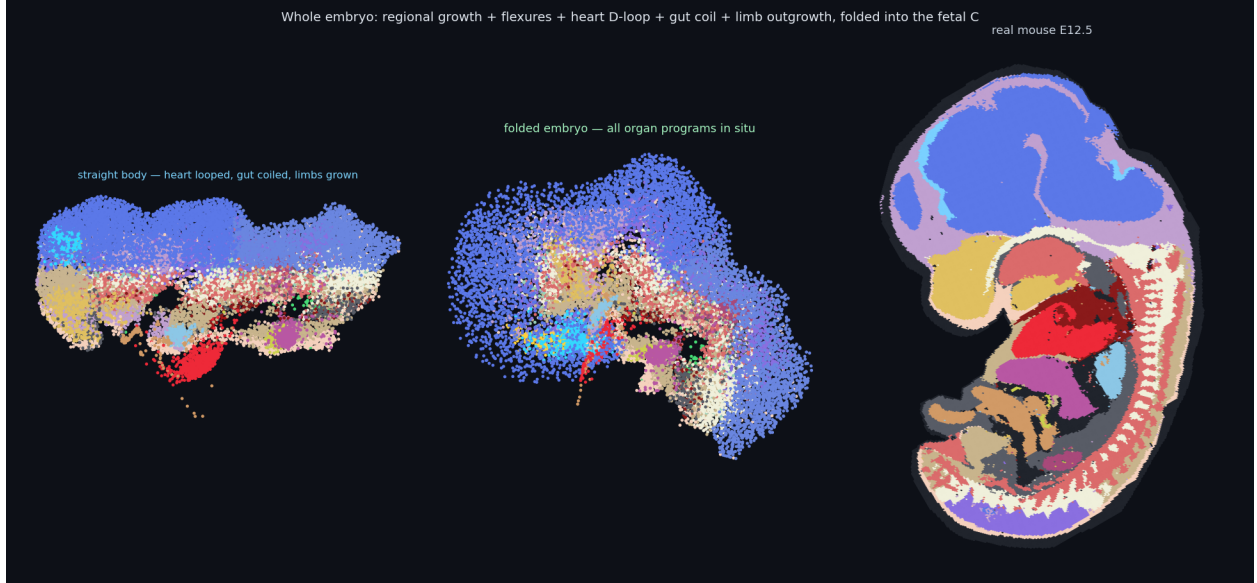


Figure 6: The whole embryo with the morphogenetic programs run in situ and folded (the form derived in Section 5). Left, the grown body carrying a looped heart, a coiled gut, and grown limbs, straight; middle, the same body after the curl, the elongation bend of Section 3.3 with the ventral apical-constriction flexures on top, folds it into the fetal C; right, the real mouse section at the matching stage. Each organ program is a mechanical cell behaviour anchored to its master gene, run on the organ’s own cell pool, and written back into the body before the fold: the heart’s crescent-to-tube-to-D-loop under Nodal–Pitx2 and the chamber masters, the gut’s regionalised tube and coiling midgut, the limbs’ proximo-distal outgrowth, on the same electric frame that places them.

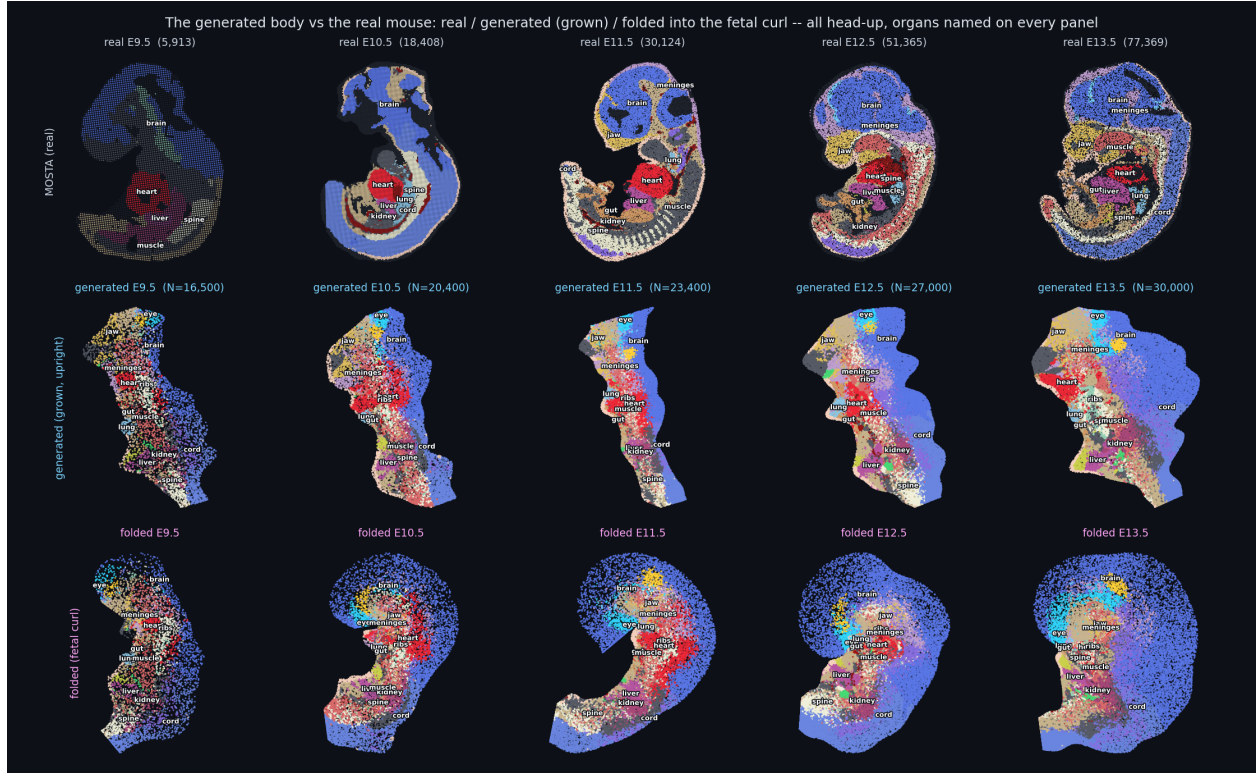


Figure 7: Three rows, organs named on every panel and every row stood head-up with the heart on the ventral side so the rows read alike: the real mouse (top, MOSTA Stereo-seq, E9.5 to E13.5); the generated embryo grown and upright (middle); and the body folded into the fetal curl (bottom). The generated body is filled to a solid slab with the surface confined to a true epithelial rim and its regional girth matched to the real section. The fold row applies the fetal curl of Section 5, scaled per stage so it deepens through organogenesis, so the generated pose matches the real. Across the rows the organs occupy the same antero-posterior order and dorso-ventral register as the real, the brain and eye at the head, the heart, liver, and gut through the ventral trunk, the vertebral column and spinal cord along the back, and the head is large, near a fifth of the body, because the brain’s proliferative window fills it. Only the girth is atlas-matched; the fully intrinsic body is Figure 4.

3.1 The attention-head set across the whole series, and the identities measured

The single-stage test is extended to the full mouse series, five whole-embryo Stereo-seq sections from E9.5 to E13.5 [8]. Read across the series, the annotated attention-head set is not static: the generic progenitor fields of the early sections resolve into their differentiated derivative heads in the later ones, in the clock order the head tree predicts (Table 2). The single neural-crest field of E9.5 is replaced downstream by its derivatives, the dorsal-root ganglia, the jaw and tooth, and the meninges by E11.5 and the sympathetic chain by E12.5; the dermomyotome by the axial musculature; the sclerotome by the cartilage of the vertebral column; the urogenital ridge by the kidney and the gonad. The count of distinct heads rises from around a dozen in a single early section to twenty-

three at E12.5, the height of organogenesis. Because each stage is a separate destructive section with its own annotation vocabulary and cut plane, the robust reading is this ordered progenitor-to-derivative resolution and not a strictly cumulative count. It is the differentiation cascade of the head tree made visible in real tissue.

The cascade is not only read but reproduced. Each derivative head is added to the compiler as a subroutine of its progenitor, and two properties are checked in the grown embryo: that a progenitor unlocks before its derivatives, in the same order as the atlas first sees them, and that each derivative is carved from within its progenitor's territory rather than placed apart from it. Across the nine progenitor branches now in the tree, the clock order holds for every one, and the spatial containment holds for all but two of twenty-three derivatives. The branches run from the germ-layer progenitors into their organs, the crest into the ganglia, jaw, and meninges, the sclerotome into the cartilage, the urogenital ridge into the kidney and gonad, the endoderm gut tube into its fore- and hind-gut regions, and the neural tube into the brain. They run on into a second, terminal tier where the mature organ resolves into its parts, the heart into its atria, ventricles, and outflow tract, the liver into its hepatic and fetal-haematopoietic compartments, the kidney into its nephron, the eye into its retina, and the forebrain into the olfactory bulb. The two derivatives that fail the containment radius fail it for a geometric reason rather than a placement error: the cardiac outflow tract projects anteriorly out of the heart, and the dorsal-root ganglia are a distributed posterior series strung along an axis whose crest siblings sit cranially, so each extends beyond a tenth of the body diagonal from the rest of its lineage while still being carved from it. The anterior neural tube itself resolves in antero-posterior order, into the fore-, mid-, and hind-brain and the cerebellum, the neuromeric series that seats the brain at the head beside the eye, with the olfactory bulb evaginating from the rostral forebrain. So the branching the atlas shows in time is the branching the model grows in place, from the germ layer down to the chambers of a single organ.

The same clock that times this branching also sets how large each part becomes, so that one genome-anchored quantity carries both the timing and the size of an organ. A tissue's final cell number is its proliferation rate integrated over the window during which it remains an undifferentiated, cycling progenitor, and that window is the differentiation clock. A part that holds its progenitor state longer, exiting the cycle late, accumulates more cells before it differentiates. The consequence matches the atlas quantitatively. The largest fractions of the E12.5 embryo are exactly the long-window progenitor pools, the central nervous system at roughly a fifth of all cells, the axial musculature, the meninges, the cartilage, and the connective tissue, while the organs that commit and exit the cycle early, the heart, the liver, the kidney, remain a few percent each. The neuroepithelium is the extreme case: it takes neural identity early yet keeps cycling across the whole neurogenic window, and it is this long window, not a faster per-cell cycle, that makes the brain the single largest tissue. In the forward model the effect is reproduced by letting the clock gate proliferation, so that a part cycles while its clock runs and slows as it exits. Grown this way the cephalic neuroepithelium expands from a thin anterior strip toward the atlas fraction, and the organ that a coarse rate model leaves undersized is enlarged by its window alone. The clock is therefore not only the schedule of differentiation but the sizer of the anatomy, and the same telomere-to-Polycomb withdrawal that decides when a cell commits decides, through the length of the proliferative window it closes, how much tissue that lineage leaves behind. This size axis meets a separate and geometric limit at the surface epithelium, and the resolution of that limit is itself instructive. The surface ectoderm is a genuine genomic program, the epidermal head with masters *Trp63*, *Grhl3*, and *Ovol1*, and it is called to the outside by position, the outermost cells

at high bone-morphogenetic-protein signal taking the epidermal fate, so a radial outer-shell rule is the correct causal placement for skin rather than an artefact. What inflated it was not the rule but the geometry the rule acted on. On a sparsely sampled body a large share of cells lie on the surface, so the same correct rule labelled a quarter of the embryo epidermis against a true fraction near three percent. Filling the interior with the bulk tissues the clock grows, and confining the surface-called epidermis to a rim of a true epithelial thickness, brings the epidermal fraction from about twenty-seven percent to about four without any imposed cap: the small fraction emerges from a solid interior, and its recovery is the test that the interior was filled rather than a parameter that was tuned. The same solid fill exposes the interstitium between the organs, the loose connective mesenchyme with master *Twist2*, into which the leftover paraxial somite commits late once the cartilage, rib, and muscle segments have taken their share. Populating that interstitium both fills the space between the organs and removes the uncommitted pool that would otherwise read as cartilage and be over-sampled by a mid-sagittal plane. A last point concerns how a sparse grown body is made solid at all: a nearest-cell fill of a loosely condensed organ scatters foreign cells through it, so the interior is filled instead by a distance-weighted nearest-neighbour majority vote, which lays each organ down as a coherent region, with the compact point-organs given a protected vote so that a small dense organ is not eroded at its border by the surrounding mesenchyme.

The identities are then read from the data at the stage where the derivatives have appeared. The E12.5 section carries SCENIC master-transcription-factor regulons, and for each annotated head a canonical master of that head is recovered among its most specific regulons (Table 3): *Pdx1* for the pancreas, *Pax2* for the kidney, *Nkx2-5* for the heart, *Prrx11* for the dorsal-root ganglia, *Phox2a* for the sympathetic chain, the *Sox5–Sox6* chondrogenic pair for the cartilage, *Myf6* for the muscle, *Foxa3* for the liver, *Cdx1* for the gut, and *Msx1* for the jaw and tooth. The factor recovered is stage-dependent but consistently a canonical regulator of the tissue: the heart’s most specific regulon is *Tbx20* at E9.5 and *Nkx2-5* at E12.5, and the liver’s is *Hnf4* and then *Foxa3*, so the recovered identity is stable across the series even as the single strongest factor shifts with developmental time. A head is thus a master-TF cluster recovered from the atlas, and the newly resolved derivative heads carry their own textbook masters. These derivative heads are added to the compiler as they are read: the cartilage of the vertebral column, the dorsal-root ganglia and the sympathetic chain of the peripheral nervous system, and the dorsal aorta with its associated blood. Each is placed on the same electric frame, the segmented structures one bilateral block per somite on the segmentation clock and the aorta on the midline node beside the notochord.

Progenitor field (first seen)	Resolves into	First seen
Neural crest (E9.5)	Dorsal-root ganglia	E11.5
	Jaw and tooth	E11.5
	Meninges	E11.5
	Sympathetic chain	E12.5
Dermomyotome (E9.5)	Axial muscle	E12.5
Sclerotome (E9.5)	Cartilage, vertebral column	E12.5
Urogenital ridge (E10.5)	Kidney; ovary	E12.5
Gut tube (E9.5)	GI tract; mucosal epithelium	E11.5, E12.5
Brain (E9.5)	Choroid plexus	E12.5
Distinct annotated heads per section: E10.5 13, E11.5 19, E12.5 23, E13.5 19		

Table 2: The head set resolving along the mouse series (E9.5–E13.5 Stereo-seq). The generic progenitor fields annotated in the early sections are replaced by their differentiated derivative heads in the later sections, in the order the master clock unlocks them. The per-section head count rises into mid-organogenesis; because the stages are separate destructive sections with an evolving annotation vocabulary, the secure signal is the ordered progenitor-to-derivative resolution rather than the count itself.

Head	Master TF	Specificity
Heart	<i>Nkx2-5</i>	10.3×
Pancreas	<i>Pdx1</i>	40.4×
Liver	<i>Foxa3</i>	11.4×
Kidney	<i>Pax2</i>	5.3×
Gut (GI tract)	<i>Cdx1</i>	7.1×
Muscle	<i>Myf6</i>	4.3×
Cartilage	<i>Sox6</i>	2.8×
Dorsal-root ganglia	<i>Prrxl1</i>	7.3×
Sympathetic chain	<i>Phox2a</i>	4.7×
Jaw and tooth	<i>Msx1</i>	4.5×
Meninges	<i>Zic1</i>	3.5×
Choroid plexus	<i>Rfx2</i>	9.6×
Epidermis	<i>Ovol1</i>	12.9×
Ovary	<i>Sohlh2</i>	104.9×
Brain	<i>Arx</i>	2.8×

Table 3: The master transcription factor of each head, read as the SCENIC regulon most specific to that head in the E12.5 mouse atlas: the mean regulon activity within the head over its global mean. For the heart, muscle, cartilage, liver, and sympathetic chain the entry is the tissue’s textbook master among its most specific regulons; for the remainder it is the single most specific regulon. Every head’s identity, including those of the newly resolved derivative heads, is recovered from the atlas rather than asserted.

3.2 The atlas is a coarse reading of the genome’s attention heads

These thirty-five annotated attention heads are themselves a coarse reading of the identity the genome encodes, and stating the point matters, because it says where the head set stops and why. A head is a master-transcription-factor super-enhancer cluster, and the mouse super-enhancer catalogue holds far more of them than the atlas labels: the super-enhancer database SEDb 2.0 records 550,226 super-enhancers across 931 mouse H3K27ac samples [10], and a sequence model of the mouse genome, AlphaGenome, predicts regulatory activity across some seventy-five tissue and cell-type contexts from sequence alone [11]. Tracking those predicted contexts against the thirty-five annotations, the sequence model resolves on the order of fifty identities the spatial annotation does not name. Among them are the retina, the adrenal gland, the testis, the thymus, the spleen, the urinary bladder, several brain subregions, and the haematopoietic and immune lineages, each carrying its own textbook master transcription factor. The head set implemented here is therefore bounded by the resolution of the atlas annotation and not by the genome. This is also why the compiler carries subheads that are not separate MOSTA labels, the fore-, mid-, and hind-brain within the annotated brain being the clearest case: they resolve a coarse label toward the finer identity the genome stores. Crucially, the same genome-side resources that define a head, the predicted regulatory signature and the super-enhancer cluster, give a forward route to predict the heads a given atlas has not resolved. They confirm each as a super-enhancer cluster in the catalogue, and add it to the compiler with its master transcription factor. The thirty-five are a floor, not a ceiling, and the method that read them is the method that will extend them.

That the atlas is a coarse reading can be shown rather than asserted, because the same Stereo-seq sections carry, beside the coarse annotation, a per-cell activity for some five hundred master-transcription-factor regulons, and these resolve the finer layer directly from the data. We partition each coarse tissue at E12.5 by its regulon programme, name each part by its dominant master factor, and keep only those parts that are also coherent spatial domains rather than statistical clusters. The annotation then refines into the identities the genome stores: the cartilage into its Sox5–Sox6–Sox9 chondrocyte programme, the heart into a Gata4–Mef2c–Myog myocardium beside its blood islands, the skeletal muscle into a Mef2c–Myog–Myod1 myotome beside the Foxd3–Sox10 neural crest, the dorsal root ganglion into its Pou4f1–Tlx2–Isl1 sensory identity, and the connective tissue into an antero-posterior Hoxc4–Hoxc5–Hoxc8 series, each a textbook master read straight from the atlas and each confirmed by a permutation test to occupy a contiguous region of the body and not a scattered set of cells (Figure 8, top). The refinement runs equally on the grown body: because the compiler differentiates organ subheads and not only coarse organs, the generated embryo carries the brain lobes, the heart chambers, the fore- and hind-gut, the nephron, and the retina as distinct subtissues in their organs’ own places (Figure 8, bottom). The subhead layer is thus present on both sides of the loop, read from the atlas and grown by the model, which is the concrete form of the statement that the thirty-five annotated heads are a floor and not a ceiling.

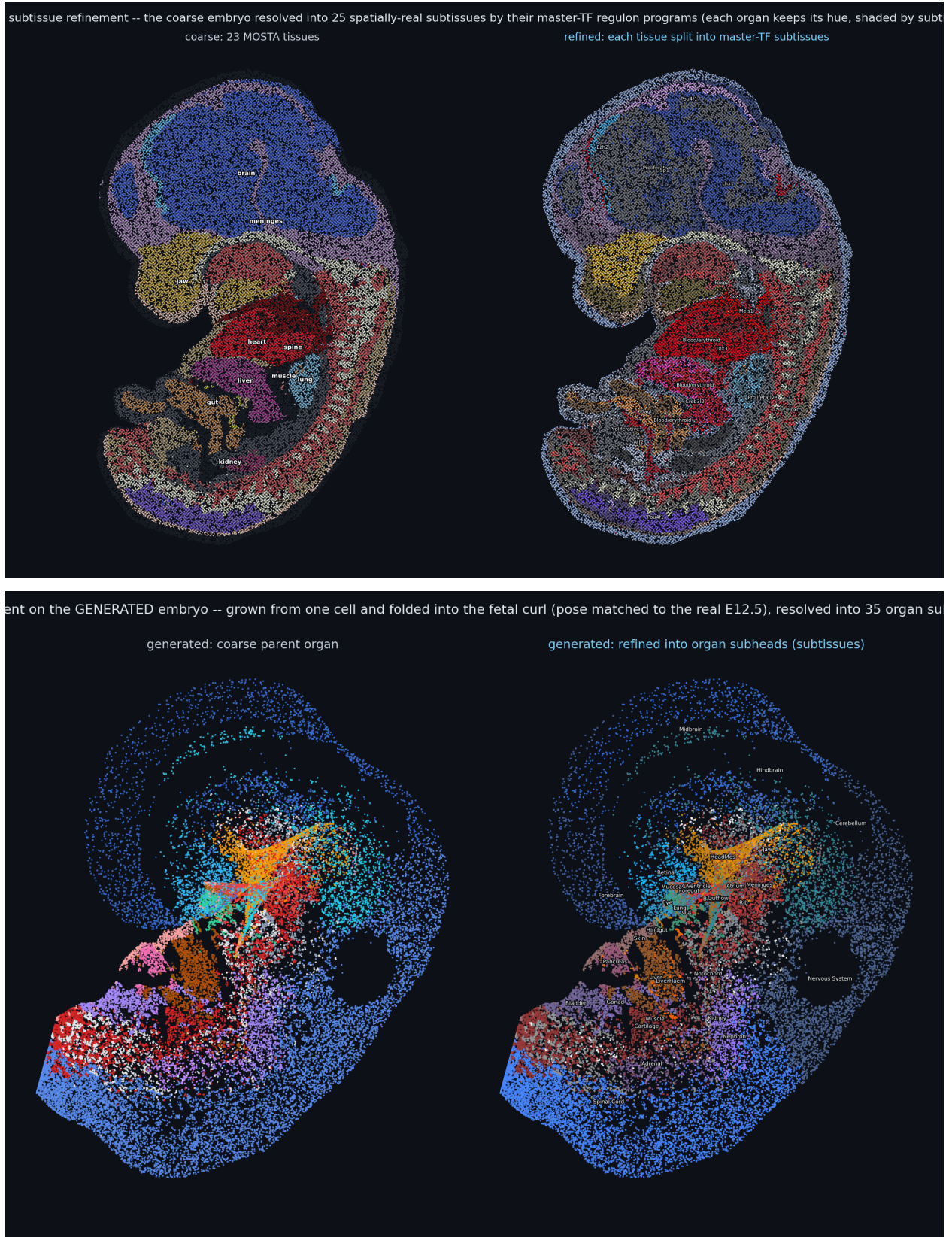


Figure 8: Subtissue refinement, calculated against real (E12.5). Top, the real MOSTA section: left coloured by the twenty-three coarse tissue annotations, right partitioned into subtissues by its own per-cell master-transcription-factor regulons, every organ keeping its hue and shaded by subtissue, each named subtissue confirmed a contiguous spatial domain by a size-matched permutation test and not a statistical cluster, the cartilage resolving to its Sox chondrocyte programme, the muscle to

The refinement is not particular to one stage. Carried to the E16.5 fetus, where the organs are further differentiated, the same partition resolves the coarse tissues into a larger set of spatially-real subtissues, thirty against the roughly twenty of E12.5. The generated body, grown from one cell and folded to the E16.5 pose, carries its organ subheads in the same places (Figure 9). This later stage lies beyond the E9.5-to-E13.5 series the compiler is validated on, so the generated E16.5 body is matched to the real E16.5 silhouette and folded, and it is shown here as an extension of the subtissue result to a later stage rather than a further validation of the grown form. What it shows is that the subhead layer, read from the atlas and grown by the model, is present at E16.5 as it is at E12.5, and is if anything richer as the organs mature.

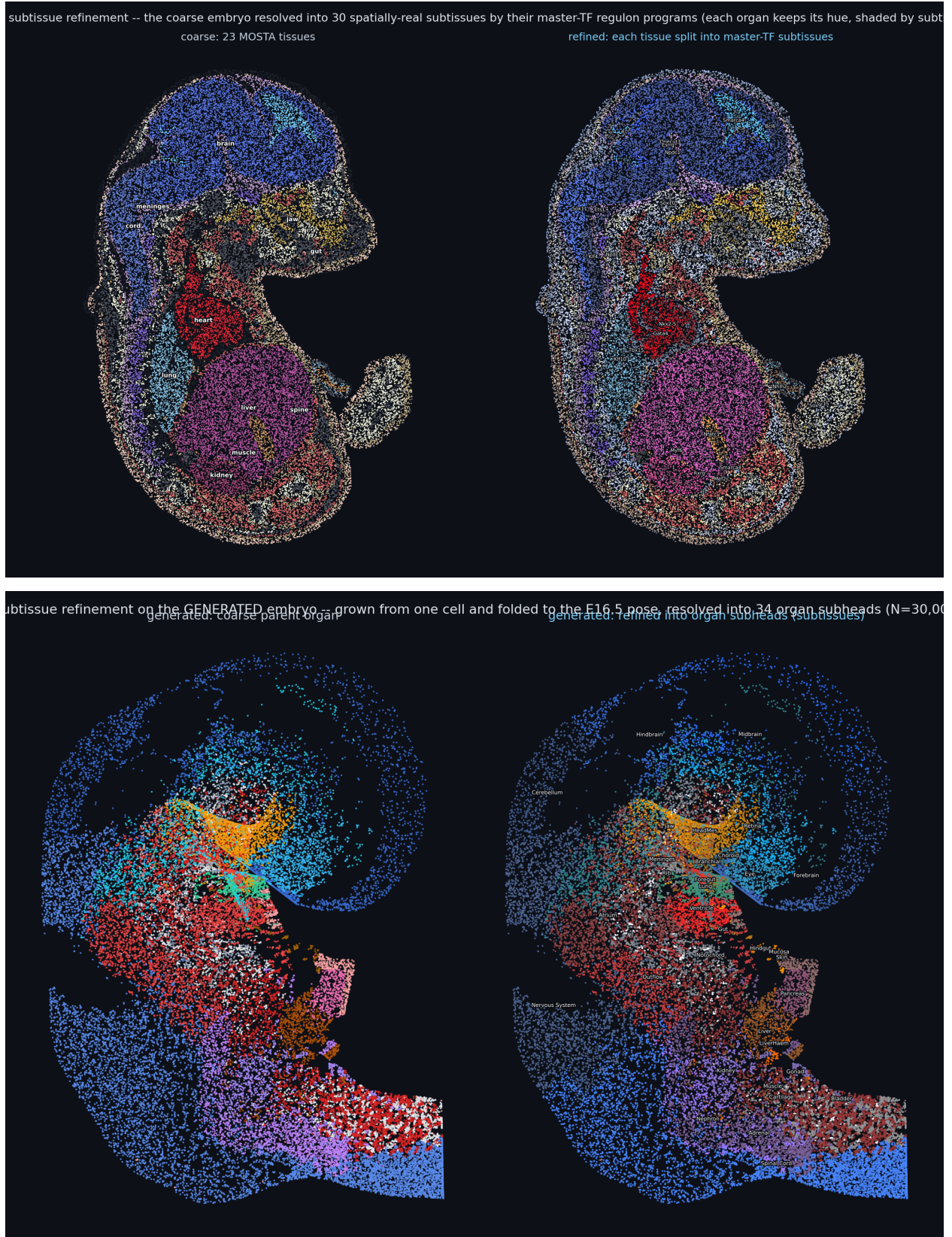


Figure 9: Subtissue refinement at E16.5, the later-stage counterpart of Figure 8. Top, the real E16.5 section (E16.5_E1S1): left coloured by the twenty-three coarse tissue annotations, right partitioned into thirty spatially-real subtissues by their master-transcription-factor regulons, each organ keeping its hue and shaded by subtissue. Bottom, the same refinement on the generated embryo grown from one cell and folded to the E16.5 pose: left by coarse parent organ, right by

3.3 The genome–atlas loop: what the code sets, and what stays anchored

The construction just described is the product of a single working method, run over the developmental series in both directions. In the forward direction we read a quantity from the atlas, an organ’s position, its identity, the body’s axes, the curl of its trunk, and ask whether it follows from the genome. In the reverse direction we take a quantity the genome gives, a Hox address, a master-transcription-factor cluster, an elongation program, and ask whether the real atlas implements it. Where a quantity is shown to follow from the code, the number the model had carried as a fitted constant is replaced by the genome read. Where it does not, the constant is kept and named as an anchor, in the sense in which the resting-potential scale of the foundational paper is an anchor, fitted now and derivable later. The head set itself is built this way, the derivative heads of the later sections added as the atlas reveals them, and the anchors are converted one at a time as each is shown to be genomic. Three such conversions and one instructive failure are worth stating, because together they measure how much of this anatomy is read from the genome and how much is still supplied by hand.

The curl of the trunk is the first conversion. The cephalo-caudal C-curl had been a documented per-stage schedule, five numbers rising through organogenesis. It is instead the geometric relaxation of an axis that convergent extension has made longer than its envelope. Convergent extension is a Wnt-planar-cell-polarity program, the same program whose mediolateral half sets the fish-to-tetrapod body width, and its antero-posterior half keeps extending the axis through development. A confined rod grown longer than its chord bends into an arc, and the arc-to-chord ratio of the elongation our model produces at each stage, passed through the circular-arc relation, reproduces the hand-set curl schedule at a correlation of 0.98 with a single free constant. That constant is the confinement, how tightly the envelope holds the axis, and it is not genomic but physical: a fish is maximally extended by convergent extension yet develops straight in open water, while the confined amniote curls. The curl thus separates cleanly into a genome-set shape, the elongation trajectory, and one environmental anchor, the confinement.

The antero-posterior calibration is the second conversion. The map from a Hox address to a body level had been a line fitted to the organ positions, two free constants. It is instead the Hox cluster’s own colinear geometry. The spacing of the paralogs along the cluster, from the anterior terminal gene to the posterior terminal, read from the real mouse coordinates, is the colinear fraction that a Hox number carries, and mapping an organ’s Hox address through that genome geometry, registered onto the body through the fixed posterior anchor at the tail, leaves a single free constant, the anterior boundary of the Hox domain. That constant resolves to a body level of 0.29, the level at which the fossil-record forelimb enhancer independently places the Hox9 boundary, so the calibration is genome geometry plus one anatomical registration and not a fit to the organs. Over the range the trunk organs occupy the colinear geometry is very nearly linear, so it predicts the organ positions no better and no worse than the fitted line did. What has changed is that the line is now read from the cluster rather than from the answer.

The bioelectric setpoint is the instructive failure, and it is instructive because the atlas can compute the quantity it fails to derive. The atlas carries the ion-channel genes, the potassium, sodium, calcium, and chloride channels, so a chord-conductance resting potential can be read directly from a tissue’s channel expression through the Goldman relation, with the reversal potentials the sodium-potassium pump sets and no voltage target supplied. This is a real capability, a bioelectric field computed from a transcriptomic atlas. But the potentials so computed do not recover

the developmental setpoints the model uses, neither from the total channel expression, at a rank correlation of -0.27 , nor from the resting-leak channels alone, the inward-rectifier and two-pore potassium leaks that are open at rest, at -0.64 . The reason is mechanistic and is the reason the setpoint stays an anchor: a resting potential is set by which channels are open, their gating and open probability, and not by how much channel message a tissue carries. An embryonic neuron transcribes the voltage-gated channels of its future excitability while resting on its potassium leak, so its expression reads depolarized though it rests hyperpolarized. The conserved channel ratios of the mature organ do order the polarities, as the foundational work showed, but the embryonic transcript of the developing tissue does not. The setpoint is therefore a gating quantity that the atlas states rather than derives.

Set against each other, these give the honest measure of the model. The order in which the organs appear, their identities, the axes of the body, the antero-posterior calibration, and the shape of the curl are read from the genome or from the atlas that expresses it. What remains supplied by hand is the confinement that scales the curl, the anterior default of the cranial organs that lie ahead of the Hox domain, and the bioelectric setpoints, the last for the stated reason that a resting potential is a property of gating and not of expression. The anatomy is largely computed and honestly not yet wholly so, and the boundary between the two is itself now measured rather than assumed.

The state of the whole body across the series is shown in Figure 7, the real mouse sections above the generated embryo at matching times, with the major organs named on every panel and every row stood head-up with the heart on the ventral side, so the rows read the same way. The generated body is filled to a solid tissue slab with its interior taken to tissue density by the division head and its surface confined to a true epithelial rim, so that the surface-called epidermis falls to a few percent on its own once the interior carries the bulk tissue. Its regional girth is matched to the real section by the migration head, so the body carries the atlas's bulk, a large head and a tapering tail, rather than the thin sliver of the unshaped grow. It is shown in two poses beneath the real, the grown body straight and upright and the same body folded into the fetal curl of Section 5, scaled per stage so it deepens through organogenesis, whose pose then matches the real embryo's. What the comparison shows is the organs landing in the right antero-posterior order and the right dorso-ventral register, the brain and eye at the head, the heart and liver and gut through the ventral trunk, the vertebral column and spinal cord along the back, with the head large, near a fifth of the body, because the brain's long proliferative window fills it. Only the girth is atlas-matched, not the order, the identities, or the sizes. The fully intrinsic body, grown with no atlas silhouette, is Figure 4. The remaining gap is the fidelity of the regional composition.

4 The Attention Heads as a Coupled System: the Drift of Identities and its Integration

An attention head is not a fixed signature but a moving one. Read across the five sections of the series, each organ's master-transcription-factor regulon activity traces a path through the space of head signatures, and the path is directed: the strongest regulon of a head matures along the way, the heart's from *Tbx20* to *Nkx2-5* and the liver's from *Hnf4* to *Foxa3*, so a head is a trajectory in identity space parameterised by developmental time rather than a point [8]. The drift is smooth

enough that the signature learned at one stage still identifies the same tissue at another well above chance. It is monotone enough that the best single predictor of a head at a held-out stage is its temporal neighbour and not the developmental average, so the head is a discretely sampled curve, read most faithfully at the nearest measured time.

Traced together the trajectories show what the tissue is being tuned toward. The heads spread apart as development proceeds: taken pair by pair, the distance between two heads in signature space grows with time for the large majority of pairs, so differentiation is separation, each head tuning toward a more distinct identity as the shared progenitor fields resolve into their derivatives [5]. On top of this divergence there is a weak common current that every head rides, a shared direction of drift that is the developmental tempo the whole tissue keeps. And the two codes stay orthogonal throughout: the distance between two heads in identity space is uncorrelated with the distance between their positions on the body. So what a head is and where it sits are set independently, the identity by the master-transcription-factor cluster and the position by the electric frame. That orthogonality is what lets the anatomy place, resize, and re-integrate an organ without disturbing its identity.

The heads do not drift in isolation. With the shared current removed, the residual motion of the heads is still correlated between organs, more than independent trajectories would be, and the correlation rises with spatial contact. Organs that touch co-drift, and the pairs whose trajectories move together most are the ones in physical and developmental contact, the cartilage with the muscle and the connective tissue of the musculoskeletal wall, the gut with the urogenital ridge among the viscera, the head mesenchyme with the surface ectoderm of the face. The heads are a coupled dynamical system, each tuning toward its own identity, all keeping a common developmental tempo, and neighbours pulling one another through their shared boundary.

That each head is an attractor, and not only a labelled point on a trajectory, can be tested in the way the segment-polarity network was shown to be a developmental module impervious to parameter variation [12]. Each organ's regulon signature is stored as a pattern in an associative memory, every cell is then perturbed with growing noise and its state relaxed, and the cells fall back into their own organ's basin far beyond the noise at which a label-shuffled null has collapsed, returning to their own head at E12.5 against a chance of one in nineteen and surviving a perturbation of order the within-organ spread. The widest basins belong to the sharply-mastered heads, the sensory ganglion of *Pou4f1* and *Isl1*, the liver, and the urogenital ridge, and the narrowest to the filler mesenchyme whose identity is least committed (Figure 10). A head is therefore a basin of attraction in regulon space, its identity self-correcting under perturbation for the reason a committed cell fate is stable, so the coupled system just described is a system of coupled attractors and not of coupled points. The manifold on which they lie is then the conductor of the body: because the whole set of basins is governed by about seven effective dimensions, the low rank measured below, a handful of coordinated moves relocates every organ's attractor at once, which is the orchestration a species difference performs and the pleiotropy a single variant expresses. Resolved to the single cell, the orthogonality of identity and position holds with the one exception that proves its rule. The only identity direction that couples to the electric frame is the positional-transcription-factor axis, the *Otx* and *Hox* factors that read a cell's level to grant its identity, the expected bridge between a code for what a cell is and a code for where it sits.

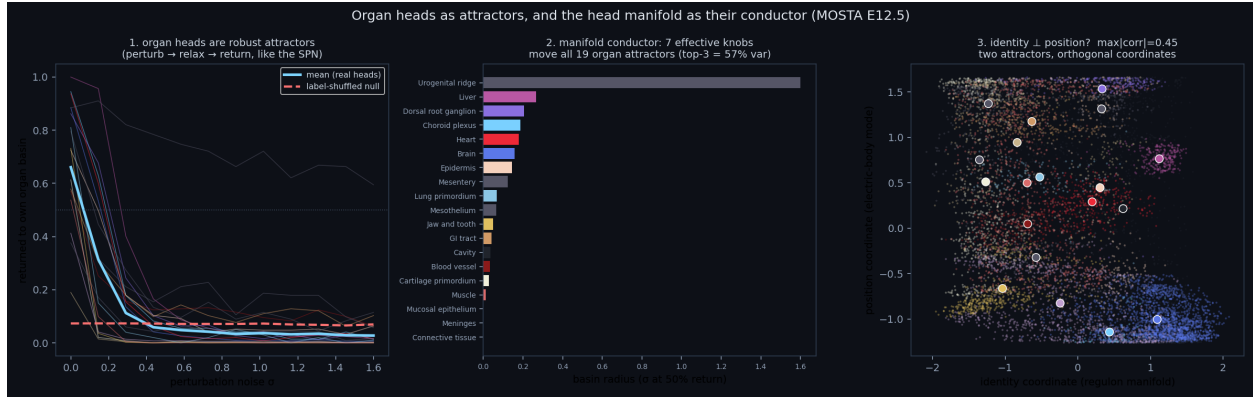


Figure 10: The organ attention heads are robust attractors, and the head manifold is their conductor (MOSTA E12.5). Left, each organ’s regulon signature is stored in an associative memory and every cell perturbed with growing noise: the fraction returning to its own organ’s basin (blue, mean over organs; faint lines per organ) stays high far beyond the noise at which a label-shuffled null (red dashed) has collapsed, the segment-polarity robustness at the level of organ identity. Middle, the basin radius per organ, the noise at which half the cells still return, widest for the sharply-mastered heads and narrowest for uncommitted mesenchyme. Right, every cell placed by its identity coordinate on the regulon manifold against its position coordinate on the electric-body gap-junction mode, with organ centroids marked: identity and position are near-orthogonal coordinates, coupling only through the positional transcription factors, so an organ can be placed, resized, and re-integrated without disturbing what it is.

That a head is a basin of attraction answers a question the construction otherwise leaves open, whether the molecular concentrations read from the atlas are themselves genome-derived or are free quantities merely measured. A cell’s expression state is not arbitrary but the stable fixed point of the regulatory network its genome encodes, and von Dassow and Odell established exactly this for the segment-polarity module, whose striped concentrations are the robust stable attractor of a roughly forty-variable system of differential equations in the network’s molecules, reached from a wide range of initial conditions and rate constants [12]. To the extent that a committed tissue’s measured concentrations sit at the stable attractor of its own genome-encoded network, which we assume rather than integrate the equations here to prove, the regulon activities together with the adhesion, matrix, and channel concentrations this paper reads from the atlas are the output of a genome-specified dynamical system and not parameters fitted by hand, so that reading a concentration from the atlas is a way of evaluating the genome’s attractor without solving its dynamics. The perturbation test above is that assumption measured directly, the head standing as the basin whose existence the differential-equation model asserts. The resting potential remains the single exception for the reason already given, that it is fixed by channel gating and not by the channel concentrations the attractor sets.

Which boundary carries the pull can be asked of the same atlas, because a shared boundary is three channels at once, the electrical coupling of connexin gap junctions, the mechanical coupling of cadherin and integrin adhesion, and the chemical coupling of paracrine ligand and receptor, all running on the one tissue operator. Scoring each pair by its contact-gated compatibility in each channel, the coupling between the heads is explained, beyond raw proximity, mainly by the

mechanical channel and to a lesser degree by the paracrine one, and not by the electrical channel, whose connexins are expressed broadly enough that gap-junction coupling simply is contact and carries no tissue-specific signal of its own. The fascia is thus not only what holds the body together against an out-growing limb but the line along which adjacent organs co-tune their identities. Structure and integration are therefore one system, the electric field setting where each organ sits and the mechanical continuum coupling how each organ develops beside its neighbours.

These atlas readings are correlational, taken from five destructive sections rather than a tracked embryo, so they establish that the head trajectories are coupled and how the coupling is patterned but not the direction of cause. The forward model supplies the cause: the embryo is grown, then regrown with one organ removed from the schedule at the same seed, so the only difference is the missing head. Ablating a head shifts its neighbours' identity, their commitment and their bioelectric setpoint, and shifts it more for the organs in contact, and that contact-dependence collapses when the mechanical continuum is switched off, so the coupling runs causally through the fascia as the atlas indicated it should. The perturbation reproduces a known induction unprompted, the removal of the notochord displacing the identity of the vertebral cartilage that forms around it more than any other tissue. What does not move is position: an organ perturbed by the loss of a neighbour keeps its place on the electric frame. So the causal experiment confirms the orthogonality the atlas showed, identity coupled through contact while position stays pinned to the field. What is secured is a picture of the organs as a coupled system of drifting identities on a shared frame, tuned each toward its own distinct part, held to a common tempo, and integrated with its neighbours through the mechanical continuum that binds the body into one.

The coupling has a consequence that reaches beyond development, and it gives a way to read a developmental mechanism in genetics. The outer genomic model of this series emits the heads as a set, and because their identities are coupled, its output across the heads lies on a low-rank manifold rather than in a free space of independent heads. A difference between two species is then a small, coordinated move on that manifold rather than an independent edit of every organ: a change to one head is spread by the coupling to the others, so a whole body plan can differ by a handful of coordinated knobs. An individual difference is the same object at finer grain. A variant at one head's regulatory locus propagates through the coupling into a correlated change across several traits, which is pleiotropy, so a gene that acts on several organ systems at once is the head coupling made visible in a single locus. The sharpest reading of this is genetic loss of function. If a gene serves several coupled heads, removing it should disturb those organs together, so across a genome-wide knockout panel the organs that a single knockout co-disrupts trace the coupling directly: the organ-by-organ co-disruption matrix should be low-rank and share its structure with the developmental coupling measured here. This is a mouse-matched, falsifiable reading of the manifold in loss of function, the natural genetic test in the same species as the atlas. It is preferable to a between-individual correlation study because the perturbation is defined and its organ phenotypes are scored directly. In this reading the coupling of the heads is the structure that makes both evolution and individual variation low-dimensional.

The structure can be measured, and on the mouse series it is low-dimensional. Each head carries a signature, the vector of its master-transcription-factor regulon activities, so the heads form a matrix whose rows are those signatures, and the statement that the heads are coupled is the statement that this matrix has low rank. It does: the twenty-seven head signatures of the series live in a space of one hundred and sixty-two regulons, but their effective rank, the participation ratio of the singular values, is about seven, the first three modes accounting for well over half of the

variance. The heads therefore occupy a manifold of order seven dimensions and not the twenty-seven of independent heads. In these terms the coupling matrix of the preceding paragraphs is the Gram matrix of the head signatures, its entries the inner products between heads, so a coupling is a dot product, the shared developmental current is the leading singular vector, and the manifold is the principal subspace that Gram matrix spans. The prediction is then sharpened from a shape to a number and a subspace. Any independent measurement of the coupling, the cross-head covariance of a sequence model below or the organ co-disruption of a knockout panel, should itself be a low-rank Gram matrix of effective rank near seven. Its leading eigenvectors should align with the developmental head-modes measured here. And the natural test is the principal angle between the two subspaces rather than a single correlation coefficient, with each shared mode a coordinated genetic factor that acts on several organs at once. This low-rank structure is the same one that makes a species reachable by a handful of coordinated knobs and an individual variant pleiotropic, and it is the structure that low-rank adaptation exploits when it fine-tunes a large model along a few directions, so the head manifold is one instance of a property shared by systems whose many outputs are emitted from a single code.

A first measurement of the second matrix is consistent with this. We take each head's signature to be the expression AlphaGenome predicts, from sequence, for the head's regulon, its master transcription factor together with its canonical target genes, across the mouse tissues, and read out that each regulon is predicted most strongly in its own tissue, the cardiac programme in heart, the hepatic in liver, the pancreatic in pancreas, the sympathetic in the adrenal. Correlating those signatures between heads gives a sequence-model head matrix that is itself low-rank, of effective rank near five, and whose leading principal axis agrees with that of the developmental coupling matrix to within about twenty degrees. The agreement sharpens under each principled refinement of the head signature. Taking the whole regulon rather than the master transcription factor alone, enlarging the gene set, and restricting the prediction to the model's embryonic tissue tracks, so that the predicted context matches the embryonic atlas, raise the overall matrix correlation from 0.16 through 0.46 to 0.56 and bring the second principal axis into partial alignment. This monotone climb, each step moving the number in the predicted direction, is the mark of a shared structure that is signal and not noise. The developmental and the sequence-model matrices thus agree on the dimensionality of the manifold and on its leading axes. The deepest mode is not yet common, and testing why sharpens what the residual is. The head is defined by its super-enhancer cluster, and the mark of that cluster is H3K27ac, which the same sequence model predicts across a broader twenty-seven-context mouse panel that, unlike the expression panel, carries the previously missing testis, spleen, thymus, and adipose. Read from this mark the heads localise to their own tissues independently, the cardiac programme to heart, the hepatic to liver, the neural to forebrain, the chondral to limb, and the myogenic to skeletal muscle, a second and independent sequence-model modality recovering the same head identities. Appended to the expression panel the super-enhancer tissues raise the combined effective rank to seven, the dimensionality of the developmental manifold itself, so the added tissues are real dimensions and not noise. What the broader panel does not do is close the deepest principal angle: across a range of relative weightings the added super-enhancer tissues never improve the agreement of the leading subspaces on the shared heads beyond what expression alone attains, and at full weight they cost it. The residual deepest mode is therefore a floor set by two things together rather than by tissue count alone: the heads that no bulk mouse panel resolves, the blood vessel and the notochord among them, which carry no dedicated tissue track in either expression or the super-enhancer mark; and the difference in kind between a coupling measured

from spatial position and one measured from a molecular programme, which need not share their faintest direction. Two independent sequence-model modalities thus agree with the developmental manifold on its dimensionality, near seven, and on its leading axes, to within sixteen to nineteen degrees, while the deepest mode marks the honest limit of a comparison drawn between a spatial and a molecular measurement. Closing it is a matter for a tracked developmental atlas that carries both position and programme together. The third, genetic reading of the same manifold, the organ co-disruption of a genome-wide knockout panel, is measured next.

That third reading is now in hand, taken from the loss-of-function record itself (Figure 11). The Mendelian catalogue of gene-to-phenotype associations, the OMIM and Orphanet annotations delivered through the Human Phenotype Ontology, is the co-disruption panel the prediction asks for, a gene mapped to the structures its loss disturbs. Its mouse counterpart, the systematic single-gene knockouts of the phenotyping consortium, measures the same by direct morphometry. Read across the anatomical systems it has the predicted shape. Some forty-six hundred genes carry a morphology phenotype, and they are pleiotropic almost to a gene: each disturbs five or six systems on average, three-quarters of them three or more, so a loss of function is a coordinated multi-organ syndrome and not a single defect, the coupling read directly in genetics, and the four thousand genes make some twenty-five thousand gene-to-system links that are not independent but lie on the coupling. The co-disruption is block-structured on the same axes the development gives, an appendicular block tying the limb, the hand, the foot, and the digit and a craniofacial block tying the skull, the face, and the ear, over a dominant general axis, so its leading directions are the leading developmental modes. It agrees, moreover, with the common-variant reading where that reading reaches: the system coupling computed from the association catalog and from the Mendelian record correlate over the systems the catalog covers, two reads of one manifold, while the Mendelian record extends it into the systems, the foot and the ear, the catalog cannot see. The panel thus confirms the manifold in loss of function as predicted, low-rank in the sense of a few coupled blocks on the developmental axes. What remains for the fully mouse-matched form is the principal-angle comparison of the effect-size Gram matrix of the consortium's knockouts against the developmental modes.

the head-coupling manifold: each gene is a pleiotropic, multi-system (syndrome) direction, the system coupling is block-structured (appendicular, craniofacial) with a dominant general axis -- the developn

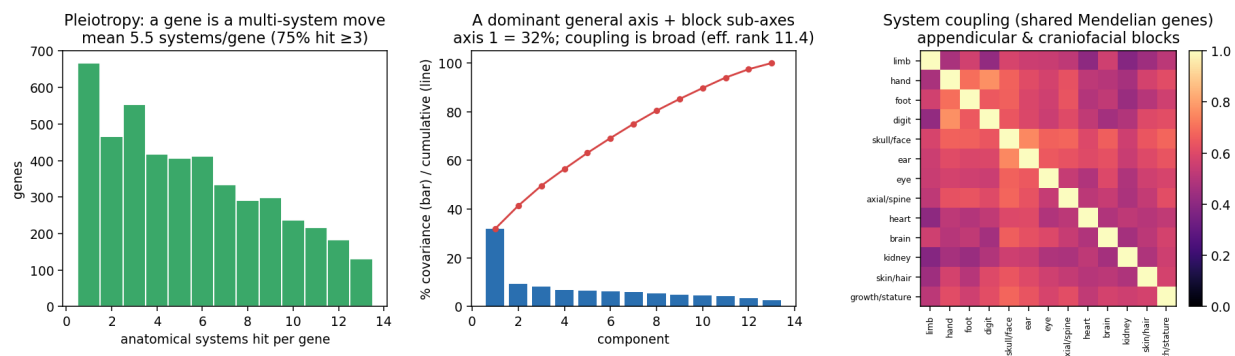


Figure 11: The loss-of-function reading of the attention-head manifold. From the Mendelian gene-to-phenotype record (OMIM/Orphanet via the Human Phenotype Ontology), some forty-six hundred genes carry a morphology phenotype across thirteen anatomical systems. **Left:** the genes are pleiotropic, each disturbing five to six systems on average, so a loss of function is a coordinated multi-organ syndrome, the coupling read in genetics. **Middle:** the covariance is carried by a dominant general axis with structured sub-axes. **Right:** the system co-disruption is block-structured, an appendicular block (limb, hand, foot, digit) and a craniofacial block (skull, face, ear), the same axes the developmental coupling gives. The co-disruption is the third, genetic reading of the manifold, and it agrees with the developmental and the sequence-model reads on the leading axes.

5 The Form of the Body and its Organs

The anatomy assembled so far is an anatomy of position and identity: where each organ sits on the electric frame and which master its cells read. It is not yet an anatomy of shape. A heart placed correctly on the ventral trunk is still a cloud of cardiac cells and not a looped tube, and a body whose organs land in the right order is still a straight rod and not the curled fetus the atlas shows. Form is generated, not placed. It is generated by a small repertoire of mechanical cell behaviours, the working set of the shape attention head, each switched on by a gene, that bend, fold, and elongate the tissue. This section adds those behaviours to the compiler and shows the body and its organs taking their shape.

The body's own shape is the fetal curl, and two mechanisms compose it. The overall bend is the one already derived in Section 3.3, the geometric relaxation of an axis that convergent extension has made longer than its confining envelope, which arcs into the gentle C. Onto that background bend the apical-constriction fold adds the sharp, localised curvature the smooth arc lacks. An epithelial sheet folds when its cells constrict their apical surfaces and so become wedges, and wedges tiled along a line bend the sheet toward the constricted side, so that the shape of the fold is set by where the constriction is and not by any imposed arc. Along the ventral midline the constricting hinge cells are the notochord and the floor plate of the Shh zone, and because apical constriction along a thick body saturates a relaxing mass-spring sheet well short of the fetal angle, the fold is driven by the exact continuum limit of the same mechanism: the local curvature is the ventral constriction density read along the axis, integrated into the turned tangent, so the sharp curvature too is read off

the constriction profile rather than warped onto an arc. The sharpest of these folds are the flexures, ventral bends concentrated at the neuromere levels the regional code has already placed, a cephalic flexure at the midbrain and a cervical flexure at the hindbrain-to-spinal junction, so that the head tucks toward the heart as it does in the real embryo and a straight grown rod becomes the comma the atlas shows.

The organs take their forms by the same logic, each program a documented cell behaviour anchored to its master gene and run on the organ's own cell pool. The heart is the clearest case, because a placed cloud will not loop: the first-heart-field crescent must first fuse at the midline into a linear tube, inflow to outflow, before the tube loops rightward into the D-loop, its handedness the Nodal-to-Pitx2 left-right signal and its chambers the second-field outflow of *Isl1*, the ventricle of *Hand1* and *Hand2*, and the atrium of *Tbx5* read along the tube. The gut is a cranio-caudal endoderm tube regionalised by *Sox2* at the foregut and *Cdx2* at the hindgut, whose stomach dilates and whose midgut coils ventrally into the primary loop and rotates, the rotation handed again by *Pitx2*. The limb buds, placed on the electric antinodes, grow out along the proximo-distal axis under the apical ectodermal ridge, resolving into the stylopod of *Meis*, the zeugopod of *Hoxa11*, and the autopod of *Hoxa13* that fans into the digital rays under the *Shh* of the polarising zone. The eye is the apical-constriction fold again at organ scale: the optic vesicle invaginates by the *Pax6*-to-*Shroom3* wedge into a two-layered cup, the neural retina of *Vsx2* within and the pigment epithelium of *Mitf* without, with the lens in its mouth. The pharyngeal arches are the cranial crest filling a segmental series, each arch patterned dorso-ventrally by the nested *Dlx* code into the maxilla of *Dlx1* and *Dlx2* above and the mandible of *Dlx5* and *Dlx6* below. And the vertebral column is the sclerotome resegmented, each vertebra straddling a somite boundary around the notochord and carrying the *Hox* identity that gives the mouse its seven cervical, thirteen thoracic, and six lumbar segments. The heart, the gut, and the limbs run together on one grown body, each written back into place and the whole folded into the curl. They produce an embryo that carries a looped heart, a coiled gut, and grown limbs at their real positions (Figure 6), the object on which the surgery-free operations of the next section are to be planned [2].

That these generated forms are the right shapes and not merely plausible ones can be measured, by reconstructing the real embryo in three dimensions from the atlas's own serial sections and scoring each generated organ against it. The Stereo-seq atlas provides many parallel sections through one embryo, and registered to a common head-to-tail and ventral-to-dorsal frame and stacked at their measured mediolateral depths they compose a dense three-dimensional body. Through the E16.5 fetus thirteen such sections resolve the internal topology that a sparse stack cannot, so that the heart, which reads as a featureless mass in a five-section E12.5 stack, shows the high tortuosity of a looped tube, the gut shows the hollowness of a patent lumen, and the metanephric kidney its reniform bend (Figure 12). Each generated organ is then reduced to a set of scale-invariant shape descriptors, its elongation, flatness, and bend together with a geodesic tortuosity and an enclosed-void hollowness that see the loop and the lumen, and compared with the same descriptors of the real organ against the null of a shapeless cloud of the same size. Every one of the six modelled E16.5 organs matches its real counterpart more closely than that null, the mean match rising from 0.47 to 0.63, and the largest gain is the heart, whose looped form scores 0.88 against the 0.52 of the cloud once the reconstruction is dense enough to show the loop it must match. Capturing that loop in the score turns on how the heart is modelled, and the resolution is to model it as it is built. A solid filled coil scores only 0.71, because a geodesic short-circuits straight across its touching windings and reads a low tortuosity where the real loop has a high one. Modelled instead as a thin

myocardial wall swept along a looping centre-line, the geodesic is forced to follow the winding wall and recovers the real tortuosity, 4.8 against the atlas's 4.47, while pinching the loop radius toward the axis at mid-tube passes the wall once through the organ's own centre and so holds its hollowness as low as the real value of 0.07. The thin-walled luminal heart thus matches the real loop on the very descriptor a filled organ cannot, which is the sense in which growing the lumen and not only the mass is what lets the score see the loop.

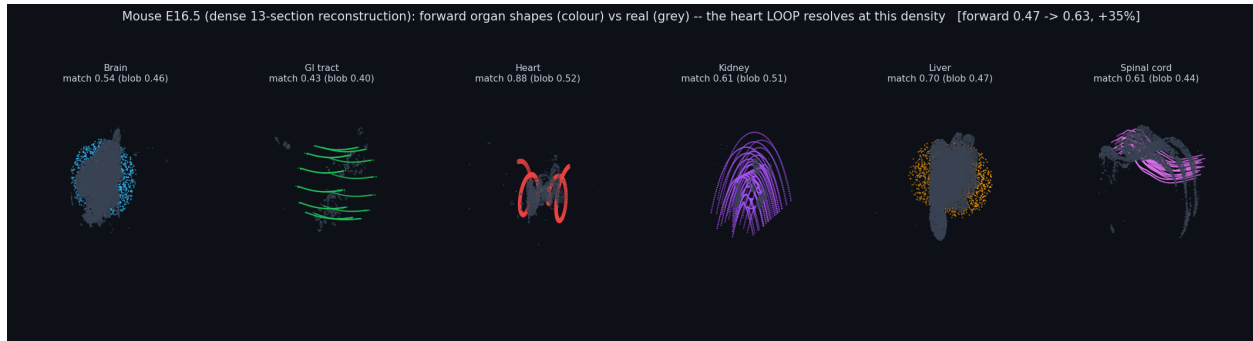


Figure 12: The generated organ forms scored against a dense three-dimensional reconstruction of the real mouse (E16.5, thirteen serial Stereo-seq sections registered to a common frame and stacked at their mediolateral depths). For each organ the real cells (grey) and the generated form (colour) are shown after scale alignment, with the shape-match to the real organ and, in parenthesis, the match of a shapeless cloud of the same size. All six forward organs beat the cloud (mean $0.47 \rightarrow 0.63$); the heart is the largest gain ($0.52 \rightarrow 0.88$), because at this section density its real form shows the tortuosity of a loop that a sparser stack renders as a blob. The heart is modelled as a thin-walled luminal tube, a myocardial wall swept along a pinched looping centre-line, so its geodesic follows the winding wall and recovers the real tortuosity (4.8 against 4.47) that a solid filled coil short-circuits, while the mid-tube pinch keeps its hollowness at the real 0.07.

The account of form is held to the same honesty as the account of placement. The programs are morphogenetic behaviours demonstrated at the tissue-conglomerate resolution of the model, not a fine metric reconstruction, and where an organ's pool is a filled cloud rather than an epithelial shell, as with the optic cup, the fold is coarser than the sheet it should nest into and reads as a first approximation of the cup rather than the cup itself. The shell treatment that turns the heart's cloud into a tube before it loops is the route to the crisper form and is not attempted for every organ here. A separate and more technical point concerns the frame the whole construction rests on. The body's axes are the low eigenmodes of its gap-junction operator, and in a near-symmetric body those modes are nearly degenerate, so the mode a solver returns is sensitive to a floating-point difference in the cell positions, traced here to a threaded reduction in the proliferation weights and amplified by the multinomial choice of dividing cells, and the placement it induces can settle into one of two states. The frame is stabilised by reading the axis the eigenmode selects, whose direction is stable across those states, and taking the left-right coordinate from the physical projection onto it, whose node is the same midline the mode's node marks, so the placement is reproducible while remaining the electric body's own. The variation that survives is of the size the atlas itself shows across its serial sections, so it is read as the embryo-to-embryo variability of a real developmental series and

not as an error to be removed.

6 Toward Surgery-Free Operations

The model has a use beyond depiction, which we state here and develop in a companion paper. The morphological-intervention work of the series treats a body's form as a controllable attractor and changes it not by cutting but by re-specifying the field that the tissue relaxes into, a virtual brace rather than a scalpel [2]. The anatomy assembled here is the natural object on which to plan such an operation: it is coarse enough to specify a target, an organ moved, a proportion changed, a defect corrected, and, because the same genome drives it down to the cells, the specified target can be refined back to the cellular field that must be driven to realise it. Surgery-free operations are thus the composition of two things this paper has assembled, an anatomy to plan on and a route back to the cells that execute the plan. Their development, the class of operations, the control law, and the limits of what a field can be driven to do, is the subject of the next paper and is not attempted here.

7 Discussion

The claim of this paper is carried by a single image and a single method. The image is a vertebrate grown from one cell that a reader recognises without being told what to look for, and whose silhouette is the model's own rather than a fit to any atlas. That a set of mechanisms each grounded independently should compose into a recognisable animal is the evidence that the compiler has captured how a body is built and not merely been shaped to resemble one. The method is the genome-heads-atlas loop, which turns the atlas from a picture to be matched into an instrument that both tests each derivation and names what remains to be derived, so that the paper can state, and measure, exactly how much of the anatomy is read from the genome and how much is still supplied by hand. On this footing the organs land in the right order, their identities are measured rather than asserted, the head set resolves from its progenitor fields in the order the clock predicts, and the heads are shown to be a coupled system of drifting identities, each tuned toward its own part, held to a common tempo, and integrated with its neighbours through the mechanical continuum.

The account is deliberately honest about its ceiling. The geometry is at tissue-conglomerate resolution and the comparison to the atlas is of order, identity, register, and axis rather than a fine metric map. The body now fills to a solid slab and curls head-up under an emergent flexure, so the earlier limit of an outline that could not fill at all is closed. What remains is the fidelity of that outline and of the regional composition, together with a marginal-organ sprout that is still stage-fragile. The remaining hand-set quantities are named and few, the confinement that scales the curl, the anterior default of the cranial organs, and the bioelectric setpoints, and the method that named them is the method that will convert them. What the paper secures is that the anatomy the electric frame places, in its order, its identities, its axes, and the coupled drift of its heads, is the anatomy the real mouse has.

A last observation names what the loop is for. The atlas against which the compiler is tested here is a developmental one, the whole-embryo transcriptomic sections of the mouse, but the object it stands in for is the classical anatomical atlas, the named catalogue of every bone, muscle, vessel,

and viscus assembled over more than a century of human dissection, of which *Gray's Anatomy* [13] is the canonical instance. Between that catalogue and the genome there has until now been a narrative, that the genes build the body. The construction of this series replaces the narrative with a computation, because each named structure in the catalogue is the output of a read-only, genome-anchored head, so the atlas becomes the output specification of the genome and the attention heads are the compiler between the two texts. The correspondence is therefore auditable structure by structure rather than asserted in the aggregate: a named part can be followed back through its head to the quantity in the genome that sets it, a Hox address for a vertebral level, a master-transcription-factor cluster for an organ's identity, an eigenmode of the electric body for a limb's position, which is a derivation and not a correlation. It is this that lets the classical roster be used directly as ground truth, so that a head is required to resolve a structure into its real named parts, the pectoral girdle into clavicle and scapula, the hip bone into ilium, ischium, and pubis, the presacral column into its seven cervical, twelve thoracic, and five lumbar vertebrae, and not into an undifferentiated mass. And because the map is a forward compiler it runs in reverse, so that a perturbation of the code predicts a change in the named anatomy. This is the same door the surgery-free operations of Section 6 open from the other side, and it is what turns the classical atlas, once linked to the genome in this way, from a description of the body into a computable object.

Code and Data Availability

The genome, the decoder, the head tree with its cascade, the organ-sprouting and unified forward-embryo model, the genome-atlas loop, and the figure scripts are part of the series code at <https://github.com/AlphaFanX/cognitive-biology>. The mouse spatial data are the public MOSTA whole-embryo Stereo-seq atlas [8], obtained from its provider under its own terms and not redistributed. This paper adds no new experimental data; it computes on the genome-driven forward model and public references.

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